



Finite element modeling of transient shear wave elastography for the cornea

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ABSTRACT

Recent advances in dynamic elastography have enabled rapid, localized, and non-invasive mechanical data acquisition of the cornea. This data opens the path to early-detection of pathologies and more accurate treatment. However, the analysis of the wave propagation is a complex mechanical problem: the cornea is a structure under pressure, with non-linear material behavior. Thus, computational analysis are needed to extract mechanical parameters from the data. In this study, we present a time-dependent finite element model for the reproduction of transient shear wave elastographic measurements in the cornea. The mechanical problem consists in a small-amplitude wave propagating in the cornea, largely deformed by intraocular pressure in physiological conditions. The model accounts for anisotropic, hyperelastic, and incompressible behavior of the cornea, as well as its accurate geometry and preloaded condition. We have implemented two different numerical approaches to solve first the static non-linear inflation of the cornea and then the linear wave propagation problem to reproduce the measurements. We investigate the impact of material anisotropy and prestress on wave propagation and demonstrate that intraocular pressure critically influences shear wave velocity. Additionally, by introducing a localized mechanical defect to simulate a pathological defect, we show that simulated shear wave can detect and quantify mechanical weaknesses, suggesting potential as a diagnostic tool to assess corneal health.

1. Introduction

Elastic wave propagation measurements represent a primary tool to assess the mechanical properties of a solid. We can think of geophysical application to reconstruct properties of terrestrial substrate from seismic waves (Zhu et al., 2012) or industrial application to estimate the health condition of a structure without damaging it (Mittra and Gopalakrishnan, 2016). In the biomedical framework, recent advancements in dynamic elastography techniques have made it possible to acquire mechanical data in a fast, localized, and non-invasive manner, positioning these methods as promising tools for in-vivo applications (Sarvazyan et al., 2011; Genisson et al., 2013). We refer in particular to methods based on a transient excitation exerted on the tissue, that generates the propagation of a high-frequency small amplitude elastic wave, captured by means of an imaging system over space and time. A mechanical and numerical model reproducing these measurements would give access to elasticity parameters that represent relevant diagnostic information for the early detection of pathologies or post-treatment care.

In this framework, the application of this imaging technique to the cornea is of particular interest from both medical and mechanical point of view. The cornea is the primary lens of the eye, with a dome-like shape and a thickness of around 500 μm . The

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cornea shape is controlled by the pressure due to the aqueous humor (inside the eye), and by the mechanical properties of the tissue due to its microstructural organization of collagen fibers. This precise organization of the collagen in the cornea guarantees the shape and transparency for a correct vision. Acquiring information in-vivo is difficult due to the reduced size, the need of non-contact techniques and the risk of compromising the patient's vision. In the past decade, steps forward have been made in the development of both exciting and imaging systems. For the cornea, we refer in particular to dynamic Optical Coherence Elastography (d-OCE) (Kirby et al., 2017; Kennedy et al., 2014; Wang and Larin, 2015; Larin and Sampson, 2017), which represents all the methods based on the use of Optical Coherence Tomography (OCT) as the non-contact, fast and high-resolution imaging system, coupled with transient excitation techniques. With this class of experimental techniques, it is possible to acquire shear wave propagation with a bandwidth up to few kHz for high-resolution and a max amplitude in the order of few μm , with no damage for the cornea (Ambrozinski et al., 2016; Singh et al., 2016; Ramier et al., 2020; Pitre et al., 2020).

From a mechanical point of view, the reproduction of d-OCE measurements in the cornea represents an interesting scenario to treat different modeling aspects and analyze the associated effect over the propagating wave. The mechanical problem to be solved is given by the elastic wave propagation in a solid with a complex geometry and a non-linear incompressible and anisotropic behavior. Furthermore, the structure is subjected to a static pressure that induces a large deformation and an associated preloaded state, affecting the dynamic response of the tissue.

There exists a large literature over mechanical models reproducing the static response of the cornea to intraocular pressure, comprised with complex constitutive law. However, to our knowledge, not the same effort has been done to model the dynamic response of the cornea, during d-OCE measurements. Most of the dynamical models are semi-analytical, based on modal analysis and simplifying assumptions. This gap between static and dynamic modeling may explain why the dynamical estimations of corneal elastic properties is few orders of magnitude below the static estimations (Pitre et al., 2020).

A reference for mathematical modeling of shear wave elastography can be found in Caforio (2019), focused on the cardiac setting. For the cornea, there are other specificities to be accounted for, such as the thin dome-like geometry, which plays a key role in the physiology of the tissue, affects the propagation of the wave as a waveguide. Similarly, it is well known that the preloaded state changes the elastic response of the material (Biot, 1940; Crutison et al., 2022), indicating the importance to including the effect of the intraocular pressure. Given that the propagating elastic wave has a small amplitude with respect to the solid, we can apply the *incremental theory* which provides the governing equation considering the propagating wave as an incremental deformation superimposed on a large deformation of the solid (Destrade and Saccomandi, 2007; Shams et al., 2011).

Biological tissues are nearly incompressible due to high water content, leading to numerical locking issues in finite element simulations (Bathe, 2006; Sussman and Bathe, 1987; Brezzi et al., 2014). In elastodynamics, explicit time discretization for large-scale problems faces severe time step restrictions from incompressibility constraints, prompting methods like selective mass scaling and conservative penalization strategies to improve stability and efficiency (Ye et al., 2017; Caforio and Imperiale, 2018). Recently, we have developed an explicit second order conservative scheme (Merlini et al., 2025) to further enhance computational efficiency without limiting the complexity of geometries.

An additional ingredient of the model is the mechanical description of the microstructure. Within the cornea, the collagen component embedded in extracellular matrix is highly organized in a hierarchical structure, with fibers mainly running along the cornea's plane and aligned in some specific directions, as we will see in more detail in Section 3. It has been possible to retrieve quantitative experimental data that describe fibers orientation and one question that may rise is to know whether the d-OCE measurements could be sensitive to the presence of the collagen network.

In order to deal with all these aspects, the outlined approach of this work starts by defining a biomechanical model of the cornea, compatible with the numerical approximation. We consider a hyperelastic nearly-incompressible anisotropic material, with an approximate description of fibers oriented distribution (Studer et al., 2010; Pinsky et al., 2005), based on the so-called *microsphere model*. By linearization, we derive the governing equations of our problem and the associated constitutive behavior of the cornea. Then, we present the numerical approaches used to solve the problem, and we analyze the response of the model to understand the effect of different modeling elements. In particular, we first assess the effect of the anisotropic model over the propagating wave, considering a linearization of the cornea response at rest. Then, we consider the effect of the prestress, first considering an analytical approximation of the cornea and then the full static inflation problem of the cornea, solved with a non-linear preconditioned conjugate gradient approach. It appears that it is the prestress that is mainly affecting the velocity of the main wave, more than the material modification during loading. As a last step, we introduce a localized mechanical defect to simulate an early stage of a pathology, the *keratoconus*. We observe that the defect can be clearly detected by the propagating shear wave, with a strong reduction of the propagation velocity that results in a deformed front wave. These results prove the applicability of the model as a diagnostic tool which, together with experimental data, could allow identifying and quantifying the severity of a mechanical weakness, in order to prevent the development of a pathological condition.

2. Problem statement

The following section introduces the notation and mathematical formulation used to model the dynamic response of the cornea, in the context of transient shear wave elastography.

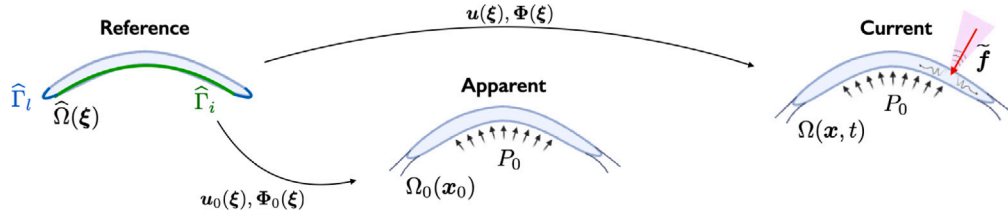


Fig. 1. Schematic representation of cornea's configurations involved in the problem. From the left: free-loaded *reference* configuration, physiological (*apparent*) configuration of the cornea fixed at the level of the limbus $\hat{\Gamma}_l$ and intraocular pressure P_0 applied on the anterior surface, and the *current* configuration of the cornea during transient shear-wave elastography. Scheme created with Biorender.com.

2.1. General problem

We indicate the domain occupied by the solid during the propagation of the wave as $\Omega(t)$, and we assume that the boundary of the cornea $\Gamma(t)$ is sufficiently smooth to have an outward normal $\mathbf{n}(t)$ defined in any point of the boundary (see Fig. 1). As the current configuration of the solid $\Omega(t)$ is unknown, it is useful to introduce a reference configuration $\hat{\Omega}$ in which we will rewrite the weak formulation of the fundamental law of dynamics. The physical points of the reference configuration of the cornea $\hat{\Omega}$ are given by the coordinate ξ , which is related to the current coordinate system $\mathbf{x}$ through the one-to-one mapping $\phi(\xi, t)$, and through which we can obtain the displacement field $\mathbf{u}$:

$$\phi : \xi \rightarrow \mathbf{x}(\xi, t) = \phi(\xi, t), \quad \phi(\hat{\Omega}) = \Omega(t); \quad \mathbf{u}(\xi, t) = \phi(\xi, t) - \xi.$$

We derive classical mechanical quantities such as the deformation gradient $\mathbf{F}(\mathbf{u}(\xi, t)) = \nabla_{\xi} \phi$, with its determinant $J(\mathbf{u}) = \det \mathbf{F}(\mathbf{u})$. We denote the Cauchy–Green tensor $\mathbf{C}$, the Green–Lagrange strain tensor $\mathbf{e}$ and its linear component $\boldsymbol{\varepsilon}$ defined as

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{e}(\mathbf{u}) = \frac{1}{2}(\mathbf{C} - \mathbf{I}), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla_{\xi} \mathbf{u} + \nabla_{\xi} \mathbf{u}^T).$$

With these definitions, we can introduce some useful relations of geometrical quantities between the current Ω and the reference configuration $\hat{\Omega}$. We have in particular that the volume element $d\Omega$ in the present configuration, can be expressed in the deformed configuration as $d\Omega = J d\hat{\Omega}$, and the surface element $ds = |\text{cof } \mathbf{F} \cdot \hat{\mathbf{n}}| d\hat{s}$, where $d\hat{s}$ is the surface element in the reference configuration, $\hat{\mathbf{n}}$ the outward normal, and $\text{cof } \mathbf{F}$ denotes the cofactor matrix of $\mathbf{F}$ i.e., $\text{cof } \mathbf{F} = J \mathbf{F}^{-T}$.

In order to have a mathematical description of the problem, the first step is to write the fundamental law of dynamics and the boundary condition of the problem and then derive the principle of virtual work, i.e. the weak formulation in the current configuration Ω . In our problem, it corresponds to the cornea during dynamic elastographic measurements. However, this configuration is unknown and changes in time and space as the wave is propagating. It is then preferable to write the problem in Lagrangian coordinates, namely in the reference configuration $\hat{\Omega}$, that is unloaded and assumed to be known. As it is a classical procedure in mechanics, we skip the passages for writing the quantities in the reference configuration, and we directly present the problem written in $\hat{\Omega}$, but we refer the reader to [Ciarlet \(1988\)](#), [Le Tallec \(1994\)](#), [Marsden and Hughes \(2012\)](#) for all the details.

For the boundary conditions, we make the following assumptions to simplify the biological complexities of the tissue. First, the cornea is subjected to the intraocular pressure (IOP), exerted by the inner fluid on the posterior surface. We take it into account as a surface load with pressure P_0 applied on $\hat{\Gamma}_i$. Second, we assume that the attachment to the sclera is infinitely rigid, so that it can be modeled by Dirichlet's conditions on $\hat{\Gamma}_l$. This also relies on the fact that techniques as d-OCE generate small amplitude waves that are fully attenuated in few wavelengths. If this is not the case, more complex models are possible, going from rotating boundary conditions ([Pandolfi and Holzapfel, 2008](#)) to simulations of the whole eye ([Sinha Roy and Dupps, 2011](#); [Kling et al., 2014](#)). Finally, model the transient pulse that generates the wave propagation as a localized body force $\mathbf{f}$.

Let us then introduce the functional spaces in which we will seek the displacement solutions

$$\mathcal{V}(\hat{\Omega}) := \{\hat{\gamma} \in H^1(\hat{\Omega})^3 \mid \hat{\gamma} = 0 \text{ in } \hat{\Gamma}_l\}, \quad \mathcal{W}(\hat{\Omega}) \subset \mathcal{V}(\hat{\Omega}).$$

with $\mathcal{W}$ the space of admissible displacements for the non-linear problem, which ensures the existence of the solution.

The non-linear wave propagation problem that we want to solve in order to reproduce dynamic shear wave elastography for the cornea reads:

For any $t \in [0, T]$, find $\mathbf{u}(t) \in \mathcal{W}(\hat{\Omega})$ such that $\forall \hat{\gamma} \in \mathcal{V}(\hat{\Omega})$

$$\int_{\hat{\Omega}} \partial_{tt} \mathbf{u} \cdot \hat{\gamma} d\hat{\Omega} + \int_{\hat{\Omega}} \boldsymbol{\Sigma} : \text{De}(\mathbf{u}) \cdot \hat{\gamma} d\hat{\Omega} = -P_0 \int_{\hat{\Gamma}_i} (\text{cof } \mathbf{F} \cdot \hat{\mathbf{n}}) \cdot \hat{\gamma} d\hat{s} + \int_{\hat{\Omega}} \mathbf{f} \cdot \hat{\gamma} d\hat{\Omega} \quad (1)$$

with initial conditions

$$\mathbf{u}(t=0) = 0, \quad \partial_t \mathbf{u}(t=0) = 0.$$

We have introduced the second Piola–Kirchhoff stress tensor $\boldsymbol{\Sigma}$, and the Gateaux derivative of the strain tensor $\mathbf{e}$ with respect to the displacements in $\mathcal{V}(\hat{\Omega})$, defined as follows:

$$\text{De}(\mathbf{u}) \cdot \mathbf{w} = \frac{1}{2} \left(\nabla_{\xi} \mathbf{w}^T \cdot \mathbf{F} + \mathbf{F}^T \cdot \nabla_{\xi} \mathbf{w} \right).$$

For hyperelastic materials, Σ is defined from the *strain energy potential* $\mathcal{W}$ through relation $\Sigma = \partial\mathcal{W}/\partial e$. The energy potential $\mathcal{W}$ is defined to model the non-linear response of the material, which for the cornea is strictly related to its complex microstructure.

2.2. Linearized wave propagation in the inflated cornea

In physiological condition the cornea is subjected to the static pressure, exerted by inner fluids on the posterior surface. During elastographic measurements, the cornea is excited by a high frequency pulse that generates small-amplitude wave propagation (Pelivanov et al., 2019; Kirby et al., 2017, 2023; Ambrozinski et al., 2016). Hence, we model the source as a transient body force δf with a small magnitude, i.e. $\delta \ll 1$.

With these loading conditions, the displacement field of problem (1) can be written as the superposition of $u_0(\xi)$, solution of a static non-linear problem given by the intraocular pressure P_0 , and $\tilde{u}(x, t)$ the transient propagating wave modeled as a small perturbation of the non-linear mechanics:

$$u(\xi, t) = u_0(\xi) + \delta \tilde{u}(\xi, t) + \mathcal{O}(\delta^2) \quad \text{with} \quad \delta \ll 1,$$

with the associated deformation gradient

$$F(\xi, t) = F_0(\xi) + \delta \nabla_\xi \tilde{u}(\xi, t) + \mathcal{O}(\delta^2) \quad \text{with} \quad F_0 := I + \nabla_\xi u_0.$$

Before linearizing, we should notice that the measured shear wave propagates in the configuration of the cornea subjected to the intraocular pressure. It is thus convenient to write the wave propagation problem in this *apparent* configuration Ω_0 (see Fig. 1), that corresponds to the cornea in physiological condition. In addition, formulating the wave problem in Ω_0 is particularly advantageous, as it allows the separation of shear and volumetric terms, enabling an effective strategy for handling the volumetric constraint in the numerical approximation.

The set of points x_0 in Ω_0 can be retrieved from the following deformation map $\phi_0(\xi)$:

$$\phi_0 : \xi \rightarrow x_0(\xi) = \phi_0(\xi) = u_0(\xi) + \xi$$

Given that, we can separate the Eq. (1) into a non-linear static problem, associated to u_0 and due to intraocular pressure P_0 , and the linearized wave propagation problem associated to $\tilde{u}$, and due to f . The full linearization procedure can be found in *Supplementary material (Sections I to III)*. The non-linear static (zeroth order) problem is:

Find $u_0 \in \mathcal{W}(\hat{\Omega})$ and $F_0 = \nabla_\xi u_0 + I$, such that $\forall \hat{\gamma} \in \mathcal{V}(\hat{\Omega})$

$$\int_{\hat{\Omega}} \Sigma_0 : \text{De}(u_0) \cdot \hat{\gamma} \, d\hat{\Omega} = -P_0 \int_{\hat{\Gamma}_i} (\text{cof } F_0 \cdot \hat{n}) \cdot \hat{\gamma} \, d\hat{s}, \quad \text{with} \quad \Sigma_0 = \Sigma(u_0). \quad (2)$$

The first order *linear dynamic problem* is: For any $t \in [0, T]$, find $\tilde{u}(t) \in \mathcal{V}(\Omega_0) := \{\gamma \in H^1(\Omega_0) \mid \hat{\gamma} = 0 \text{ in } \hat{\Gamma}_l\}$, such that $\forall \gamma \in \mathcal{V}(\Omega_0)$

$$\begin{aligned} & \int_{\Omega_0} \partial_t \tilde{u} \cdot \gamma \, d\Omega_0 + \int_{\Omega_0} \varepsilon_{x_0}(\tilde{u}) : \mathbb{C}_0 : \varepsilon_{x_0}(\gamma) \, d\Omega_0 \\ & + \int_{\Omega_0} \tilde{p} \, \text{div}_{x_0} \gamma \, d\Omega_0 + \int_{\Omega_0} \nabla_{x_0} \tilde{u} \sigma_0 : \nabla_{x_0} \gamma \, d\Omega_0 \\ & = \int_{\Omega_0} f \cdot \gamma \, d\Omega_0 - P_0 \int_{\Gamma_{0,l}} \left(\text{div}_{x_0}(\tilde{u}) I - F_0^{-T} \cdot \nabla_{x_0} \tilde{u} \cdot F_0 \right) \cdot n \cdot \gamma \, ds \end{aligned} \quad (3)$$

where the subscript $\cdot_{x_0}$ indicates that the quantity is defined with respect to the apparent configuration Ω_0 . The 4th-order tensor $\mathbb{C}_0$ and the scalar function $\tilde{p}$ refer to the shear and volumetric component of the linearized stress tensor $\tilde{\Sigma}$ in the loaded configuration Ω_0 . We refer to them as the *shear and volumetric material stiffness*, as it describes the behavior of the material linearized around the solution u_0 . The term σ_0 denotes the *prestress* induced by the intraocular pressure, and we refer to the associated bilinear form as the *geometric stiffness*. This separation allows to have a physical interpretation of the different terms and distinguish what depends on the material behavior and what depends on the prestress within our structure.

The first difficulty that we encounter in this formulation is related to the linearized surface term given by the pressure P_0 , which results from the linearization of $\text{cof } F$. (see *Supplementary material (Section III)* for details). Unfortunately, it corresponds to a non-symmetric bilinear form, preventing us from employing several tools to guarantee the stability and the well-posedness of the linear wave propagation problem. We are going to neglect the perturbation of the pressure load in the rest of the work. This is supported by the fact that the cornea is a rather thin structure (the ratio thickness h over curvature radius R is approximatively 0.07). Therefore, the main stress is the one in the in-plane direction, which has a magnitude significantly higher than the pressure: $\|\sigma_0\| \approx P_0 R/h \gg P_0$.

In addition, the cornea is approximated as nearly-incompressible, which limits the efficiency of classical explicit dynamic solvers. Thus, a dedicated numerical approach is needed (Merlini et al., 2025). As shown later, the scalar function $\tilde{p}$ reduces to $\tilde{p} = \lambda \, \text{div}_{x_0} \tilde{u}$ with λ a positive constant taken very large and consistent with the bulk modulus. We then rewrite the nearly-incompressible linear wave propagation problem with the following *mixed formulation*

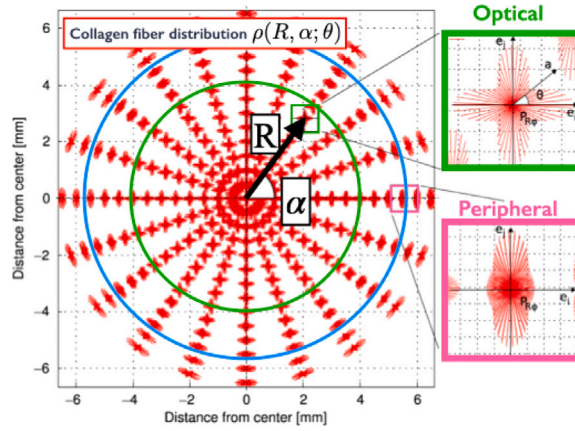


Fig. 2. Angular fiber distribution $\rho(\xi, \theta)$ proposed by Studer et al. (2010) as an approximation of the data obtained by Meek and Boote (2009). The blue circle delimits the cornea and the green circle separates the optical zone from the peripheral one. Image taken from (Studer et al., 2010).

For any $t \in [0, T]$, find $(\tilde{\mathbf{u}}(t), \tilde{p}(t)) \in \mathcal{V}(\Omega_0) \times \mathcal{L}(\Omega_0)$, such that $\forall (\gamma, q) \in \mathcal{V}(\Omega_0) \times \mathcal{L}(\Omega_0)$

$$\begin{cases} \int_{\Omega_0} \partial_{tt} \tilde{\mathbf{u}} \cdot \gamma \, d\Omega_0 + \int_{\Omega_0} \epsilon_{x_0}(\tilde{\mathbf{u}}) : \mathbb{C}_0 : \epsilon_{x_0}(\gamma) \, d\Omega_0 + \int_{\Omega_0} \tilde{p} \operatorname{div}_{x_0} \gamma \, d\Omega_0 \\ \quad + \int_{\Omega_0} \nabla_{x_0} \tilde{\mathbf{u}} \sigma_0 : \nabla_{x_0} \gamma \, d\Omega_0 = \int_{\Omega_0} \mathbf{f} \cdot \gamma \, d\Omega_0 \\ \int_{\Omega_0} q \operatorname{div} \tilde{\mathbf{u}} \, d\Omega_0 = \lambda^{-1} \int_{\Omega_0} \tilde{p} q \, d\Omega_0 \end{cases} \quad (4)$$

with initial conditions: $\tilde{\mathbf{u}}(0) = 0$, $\partial_t \tilde{\mathbf{u}}(0) = 0$. Notice that it can also be seen as a penalized formulation of a pure incompressible problem with $\tilde{p}$ defined as a Lagrange multiplier.

3. Material model of the cornea

Up to now, we have presented a general problem of small amplitude elastic wave propagation in a nearly-incompressible hyperelastic solid loaded by a static pressure term. Let us now apply the general formulation to the specific case of the cornea, starting from the definition of its macroscopic behavior, that is strictly related to its microstructure.

3.1. Material behavior

The cornea is a thin tissue organized in a layered structure, where the *stroma* is the principal layer as it occupies the 90% of the thickness. This stromal layer is characterized by the presence of a highly organized, stiff, collagen network (Komai and Ushiki, 1991). For these reasons, the material behavior of the cornea is usually described from the mechanical properties of the stroma, neglecting the contribution of the other layers.

The stroma has a plywood-like structure, with between 200 and 500 lamellae (Meek and Knupp, 2015), where collagen fibrils are almost crystallized along lamellar direction and surrounded by extracellular matrix. Thanks to X-ray scattering technique (Boote et al., 2006; Meek and Boote, 2009; Keith M. Meek, 1987; Aghamohammadzadeh et al., 2004), it was possible to retrieve quantitative information about the collagen arrangement. It has been observed that in the central part of the cornea, lamellae are mainly oriented in the naso-temporal (horizontal) direction and in the inferior-superior (vertical) direction, as shown in Fig. 2, where an analytical approximation of the X-ray scattering data is given with polar plots at all locations. As we move towards the sclera, collagen fibers tend to be mainly aligned in the circumferential direction. The signal is also characterized by a constant component referred to the isotropic part of the material, such as randomly distributed fibers and extracellular matrix. In addition, as most biological tissues, the stroma is characterized by high hydration, which is generally associated to nearly incompressible behavior.

In order to reproduce its mechanical response, we model the cornea as a hyperelastic anisotropic nearly-incompressible material. We define the strain energy function as the sum of three independent components

$$\mathcal{W} = \mathcal{W}^{iso} + \mathcal{W}^{vol} + \mathcal{W}^{ani}$$

where $\mathcal{W}^{iso}$ represents the isotropic contribution of the extracellular matrix and randomly oriented fibers. In this work, we use a classical Neo-hookean functional

$$\mathcal{W}^{iso} = \frac{\mu}{2} (J_1 - 3) \quad (5)$$

where μ is a material parameter. Instead of I_1 , we use the reduced-invariant $J_1 = I_1 I_3^{-1/3}$, which allows accounting for the energy associated only to the deviatoric deformation of the isotropic component. For a detailed discussion over the use of reduced invariants for nearly-incompressible solids, we suggest the reader to look at [Chen and Pan \(1996\)](#), [Penn \(1970\)](#). The term $\mathcal{W}^{vol}$ accounts for the nearly-incompressibility of the tissue, penalizing volumetric deformation, i.e. enforcing $J = \det F$ to be close to 1. Here we take

$$\mathcal{W}^{vol} = K(J - 1)^2 \quad (6)$$

where K is the bulk modulus, a positive constant taken much larger than the other material parameters.

Finally, the effect of the collagen network is given by the last term $\mathcal{W}^{ani}$, that will depend on the kinematics of the fibers and on their arrangement within the solid. In general, mechanical models of fibrous biological tissue should account for a certain degree of angular dispersion of fibers with respect to principal directions, in order to have a realistic mechanical response ([Holzapfel et al., 2019](#)). For this type of materials, the anisotropic component of the deformation energy density can be written in the following form

$$\mathcal{W}^{ani}(\xi, F) = \frac{1}{4\pi} \int_{\omega} \rho(\xi, \hat{r}) \mathcal{W}^f(i_4(F, \hat{r})) d\omega, \quad (7)$$

where $\rho(\xi, \hat{r})$ denotes a space-dependent density function describing fibers arrangement in the reference configuration. The distribution ρ is evaluated with respect to the general direction $\hat{r}(\phi, \theta)$, where θ is the in-plane angle, and ϕ out-of-plane angle. The potential $\mathcal{W}^f(i_4(F, \hat{r}))$ accounts for the hyperelastic response of collagen fibers through the pseudo-invariant $i_4(F, \hat{r})$, which describes fibers kinematics along direction $\hat{r}$. The integral is computed over the unit sphere ω with $d\omega = \sin \phi d\phi d\theta$, for which we refer to this model as the *microsphere model*. This approach has been introduced in [Lanir \(1983\)](#), then extended in [Gasser et al. \(2006\)](#) for arteries, and it has been applied to different types of tissues such as skin [Annaiidh et al. \(2012\)](#), [Joodaki and Panzer \(2018\)](#), tendon ([Fang and Lake, 2016](#)) and cornea ([Pandolfi and Vasta, 2012](#); [Pandolfi and Manganiello, 2006](#); [Nguyen and Boyce, 2011](#); [Nguyen et al., 2008](#); [Simonini and Pandolfi, 2015](#)). In the framework of cornea modeling, the quantitative data obtained with X-ray technique in [Aghamohammadzadeh et al. \(2004\)](#) can be used to directly retrieve parameters of in-plane fiber angular distribution ρ . We cite in particular the work of [Giraudet et al. \(2022\)](#), whose model accounts for two families of lamellae with the associated directional density distributions. While these models approximate static or quasi-static problems, here our case of interest is the reproduction of measurements based on transient elastic wave propagation. The discontinuity and noise related to experimental data would generate numerical and artificial reflections of waves, for which this approach cannot be applied.

In order to overcome this limitation, we use the continuous analytical approximation of fiber orientation and distribution *in the plane* of the cornea, that was originally proposed in [Pinsky et al. \(2005\)](#) and then expanded by [Studer et al. \(2010\)](#). Let us introduce (R, α) as the 2D in-plane polar coordinate of the flattened cornea. We also consider that in-plane, the direction $\hat{r}$ is only defined by the angle θ . Then, the angular fiber distribution is given by a linear combination of two π -periodic functions, $\rho_1, \rho_2 \in C^\infty$, such that

$$\rho(R, \alpha; \theta) = \beta(R)\rho_1(\theta) + (1 - \beta(R))\rho_2(\theta - \alpha), \quad (8)$$

with

$$\begin{cases} \rho_1(\theta) = \cos^{2n}(\theta) + \sin^{2n}(\theta) + c_1 \\ \rho_2(\theta - \alpha) = \sin^{2n}(\theta - \alpha) + c_2, \end{cases} \quad (9)$$

and

$$\beta(R) = \begin{cases} 1.0 & R < R_{opt} \\ -\frac{1}{2} \cos\left(\frac{\pi}{R_{max}-R_{opt}}(R_{max}-R)\right) + \frac{1}{2} & R_{opt} \leq R \leq R_{max}. \end{cases} \quad (10)$$

The first component ρ_1 describes the fiber distribution in the optical (or central) zone, in which we distinguish two perpendicular directions, naso-temporal and inferior-superior, while in the peripheral zone fibers are aligned in circumferential direction ([Studer et al., 2010](#)).

Experimental evidence ([Radner et al., 1998](#); [Jester et al., 2010](#); [Raoux et al., 2023](#)) has shown that fibers exhibit also a depth-dependent orientation distribution through the cross-section of the cornea, with highly intertwined lamellae as we get closer to the anterior surface. However, this distribution concerns one-third of the thickness, while in the rest of the cross-section lamellae lie mostly parallel to the in-plane surface. For this reason, in our model we consider only the *in-plane* angular fiber distribution, taken uniform through the thickness.

The constants c_1 and c_2 are defined to normalize the distribution, such that

$$\frac{1}{\pi} \int_0^\pi \rho(\theta) d\theta = 1,$$

with $n = 4$ and resulting constants are $c_1 = 0.453$ and $c_2 = 0.727$.

For the non-linear behavior of fibers, we use a non-linear exponential law defined as follows:

$$\mathcal{W}^f(F, \theta) = \frac{k_1}{2k_2} \left[e^{k_2 \left(\sqrt{i_4(F, \theta)} - 1 \right)^2} - 1 \right], \quad (11)$$

where k_1 describes the stiffness of collagen fibers and k_2 is a dimensionless parameter that helps us to describe the activation of fibers at a certain level of stretch. The non-linear potential for collagen fibers Eq. (11) is obtained from a regularization of the potential $\mathcal{W}^f$ proposed in Giraudet et al. (2022), based on the critical stretch of collagen fibers at which collagen fibers start to exert their tensile strength. Here, we adapt the model for the study of linear wave propagation. The pseudo-invariant describing the kinematics of the fibrils is given by the following relation

$$i_4(\mathbf{F}, \theta) = \hat{\mathbf{r}}(\theta) \cdot \mathbf{C} \cdot \hat{\mathbf{r}}(\theta) \quad (12)$$

where $\hat{\mathbf{r}}$ corresponds to the general direction lying on the tangent plane of the cornea surface at point ξ .

A common choice when modeling hyperelastic soft tissues is to use the reduced invariant $J_4 = J^{-2/3} I_4$ for the anisotropic component, in order to split the deviatoric and volumetric component of the potential. A recent study (Jin et al., 2022) shows how this latter choice introduce a *locking* effect when modeling non-linear anisotropic biomaterials. In particular, it is observed that a solid with the hyperelastic anisotropic behavior based on J_4 subjected to an uniaxial stretch, will have a variation of the internal pressure that strongly depends on the ratio between the mechanical parameters of the isotropic and anisotropic component, which here would be μ and k_1 , increasing with the amount of stretch applied. Furthermore, the Jacobian determinant $J = \det \mathbf{F}$ will strongly vary with both material parameters ratio and stretch, violating the nearly-incompressibility of the material. Instead, when using the pseudo-invariant I_4 , the internal pressure does not depend on material parameters of isotropic and volumetric potentials and the incompressibility is preserved.

3.2. Material response for static and dynamic responses

In order to solve the linearized wave propagation problem ??, we need to separate the material behavior into zeroth and first orders contribution. The procedure is done for the specific case of the cornea but can be easily applied to other materials. It follows a classical linearization: first, for each component of the strain energy density ($\mathcal{W}^{iso}$, $\mathcal{W}^{vol}$, $\mathcal{W}^{ani}$), we derive the stress tensor Σ . We obtain $\Sigma = \Sigma^{iso} + \Sigma^{vol} + \Sigma^{ani}$, with

$$\begin{aligned} \Sigma^{iso} &= \mu \left(I_3^{-\frac{1}{3}} \mathbf{I} - \frac{1}{3} J_1 \mathbf{C}^{-1} \right) \\ \Sigma^{vol} &= 2K(J-1)J\mathbf{C}^{-1}. \\ \Sigma^{ani} &= \frac{1}{\pi} \int_0^\pi \rho(\xi, \theta) 2 \frac{\partial \mathcal{W}^f}{\partial i_4} \hat{\mathbf{r}}(\theta) \otimes \hat{\mathbf{r}}(\theta) d\theta \\ \text{where} \quad 2 \frac{\partial \mathcal{W}^f}{\partial i_4} &= k_1 \left(1 - \frac{1}{\sqrt{i_4}} \right) e^{k_2(\sqrt{i_4}-1)^2} \end{aligned}$$

With these relations we can solve the static nonlinear problem (2).

Then, we linearize Σ by separating the terms related to the non-linear static solution $\mathbf{u}_0$ from those of the first-order dynamic contribution $\tilde{\mathbf{u}}$. Finally, we reformulate the quantities in the apparent configuration Ω_0 , to retrieve the components of formulation (4). Since the linearization approach is well-known, we only presents the final results, but the interested reader can find the details in *Supplementary material (Section IV)*. We obtain:

- the *shear material stiffness*, associated to the linearized shear response of the cornea. Here, we present it decomposed in the contribution from the isotropic, volumetric and anisotropic strain energy density:

$$\begin{aligned} \int_{\Omega_0} \epsilon_{x_0}(\tilde{\mathbf{u}}) : \mathbb{C}_0 : \epsilon_{x_0}(\gamma) d\Omega_0 &= \\ \int_{\Omega_0} \epsilon_{x_0}(\tilde{\mathbf{u}}) : (\mathbb{C}_0^{iso} + \mathbb{C}_0^{vol} + \mathbb{C}_0^{ani}) : \epsilon_{x_0}(\gamma) d\Omega_0 & \end{aligned} \quad (13)$$

where

$$\begin{aligned} \mathbb{C}_0^{iso} : \epsilon_{x_0}(\tilde{\mathbf{u}}) &= \frac{2}{3} \mu J_0^{-\frac{5}{3}} \left[I_{1,0} \epsilon_{x_0}(\tilde{\mathbf{u}}) \right. \\ &\quad \left. - \left((\mathbf{I} : \epsilon_{x_0}(\tilde{\mathbf{u}})) \mathbf{F}_0 \cdot \mathbf{F}_0^T + (\mathbf{F}_0 \cdot \mathbf{F}_0^T : \epsilon_{x_0}(\tilde{\mathbf{u}})) \mathbf{I} \right) \right] \end{aligned} \quad (14)$$

$$\mathbb{C}_0^{vol} : \epsilon_{x_0}(\tilde{\mathbf{u}}) = -4K(J_0-1)\epsilon_{x_0}(\tilde{\mathbf{u}}) \quad (15)$$

$$\mathbb{C}_0^{ani} : \epsilon_{x_0}(\tilde{\mathbf{u}}) = \frac{1}{J_0} \left[\frac{1}{\pi} \int_0^\pi 4 \rho(R, \varphi; \theta) \frac{\partial^2 \mathcal{W}^f}{\partial i_4^2} \Big|_{i_{4,0}} \tilde{\mathbb{R}}(\theta) d\theta \right] : \epsilon_{x_0}(\tilde{\mathbf{u}}) \quad (16)$$

with $\tilde{\mathbb{R}}(\theta) = \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \otimes \mathbf{r}(\theta)$

$$\frac{\partial^2 \mathcal{W}^f}{\partial i_4^2} \Big|_{i_{4,0}} = \frac{k_1}{2i_{4,0}} \left(\frac{1}{2\sqrt{i_{4,0}}} + k_2(\sqrt{i_{4,0}}-1)^2 \right) e^{k_2(\sqrt{i_{4,0}}-1)^2}$$

- the *volumetric material stiffness* associated to volumetric deformations

$$\int_{\Omega_0} \tilde{p} \operatorname{div}_{x_0} \gamma \, d\Omega_0 \quad \text{with} \quad \tilde{p} = \left(2K \frac{(2J_0 - 1)}{J_0} - \frac{2}{9} \mu J_0^{-\frac{5}{3}} I_{1,0} \right) \operatorname{div}_{x_0} (\tilde{\mathbf{u}}) \quad (17)$$

- the *geometric stiffness*, depending on the prestress $\sigma_0 = J_0^{-1} F_0 \cdot \Sigma_0 \cdot F_0^T$ (the Cauchy stress tensor in Ω_0), such that

$$\int_{\Omega_0} \nabla_{x_0} \tilde{\mathbf{u}} \sigma_0 : \nabla_{x_0} \gamma \, d\Omega_0 = \int_{\Omega_0} \nabla_{x_0} \tilde{\mathbf{u}} (\sigma_0^{iso} + \sigma_0^{vol} + \sigma_0^{ani}) : \nabla_{x_0} \gamma \, d\Omega_0 \quad (18)$$

with

$$\sigma_0^{iso} = \mu J_0^{-5/3} (F_0 \cdot F_0^T - I) \quad (19)$$

$$\sigma_0^{vol} = 2K (J_0 - 1) I \quad (20)$$

$$\sigma_0^{ani} = \frac{1}{J_0} \frac{1}{\pi} \int_0^\pi 2\rho(R, \varphi; \theta) \left. \frac{\partial \mathcal{W}^f}{\partial i_4} \right|_{i_{4,0}} \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \, d\theta \quad (21)$$

$$\text{where} \quad 2 \left. \frac{\partial \mathcal{W}^f}{\partial i_4} \right|_{i_{4,0}} = k_1 \left(1 - \frac{1}{\sqrt{i_{4,0}}} \right) e^{k_2 (\sqrt{i_{4,0}} - 1)^2} \quad (22)$$

Notice that the notation $(\cdot)_0$ denotes that the quantity is evaluated in $\mathbf{u}_0$.

From these results, there are some aspects that we would like to point out. First, the linearization of both isotropic and volumetric stress tensors contribute to both shear and volumetric material stiffness (in Eqs. (14), (15), and Eq. (17)). This coupling is due to the choice of using reduced invariants J_i for the potentials (5) and (6), that is fundamental to ensure the stability of the wave propagation problem. For the anisotropic component of the potential (7) we have used instead the pseudo-invariant i_4 , such that the linearization of the associated stress tensor results only in a shear material stiffness term.

In addition, we should notice that the static internal pressure $p_0 := -2K(J_0 - 1)$, computed from the static non-linear problem, is a source of instability. This is a well-known problem in literature (Bathe, 2006; Sussman and Bathe, 1987). It is then important to have a numerical strategy that ensures $J_0 \approx 1$, when enforcing nearly incompressibility, and to verify the stiffness coercivity of the wave propagation problem.

Another comment concerns the linearization of the anisotropic component and the definition of vector $\mathbf{r}(\theta)$ in Eqs. (16), (21). From the formulation of the quantities in Ω_0 , we obtain $\mathbf{r}(\theta) = F_0 \cdot \hat{\mathbf{r}}(\theta)$, which denotes the general direction tangent to the surface of the cornea subjected to intraocular pressure. The fourth-order tensor $\mathbb{R}(\theta)$ describes fibers' deformation with respect to the apparent configuration Ω_0 . Furthermore, it has major and minor symmetries (Eq. (16)) by construction, such that the full tensor $\mathbb{C}_0$ has only 21 independent components.

Remark 1. Since the enforcement of nearly-incompressibility is obtained with the bulk modulus K taken much larger than the other material parameters and the quantity J_0 from the non-linear static problem will be close to 1, it is reasonable to compute the volumetric term (17) with the following approximation

$$\tilde{p} = \lambda \operatorname{div}_{x_0} \tilde{\mathbf{u}} \quad \text{with} \quad \lambda \simeq 2K.$$

In addition, the isotropic component of the shear stiffness reduces to

$$\mathbb{C}_0^{iso} : \epsilon_{x_0}(\tilde{\mathbf{u}}) = \frac{2}{3} \mu J_0^{-\frac{5}{3}} I_{1,0} \epsilon_{x_0}(\tilde{\mathbf{u}})$$

considering that the quantity $\operatorname{div}_{x_0} \tilde{\mathbf{u}}$ will tend to zero as the cornea is nearly-incompressible, and with a change of definition of the pressure $\tilde{p}$.

4. Numerical model of elastic wave propagation in the cornea

4.1. Numerical approaches

Now that we have derived a formulation of the problem with all the associated mechanical quantities, let us present the finite element model that has been developed for the reproduction of dynamic elastographic measurements of the cornea. Since the dynamic response of the cornea is approximated as a small amplitude wave propagation superimposed to a static finite deformation, we have to consider a different approach to obtain the two components of the solution.

The static non-linear problem (2) is solved by means of a classical iterative method, a non-linear conjugate gradient method with a Laplacian preconditioner (Exl et al., 2019). In practice, the static solution is found from a minimization process of the non-linear potential associated to the weak formulation. The method is acceptable for mild homogeneous deformation such as the inflation of the cornea in physiological condition, and allows an easy connection with the solver implemented for the linearized wave propagation problem. Some details about the solver are given in the *Supplementary material (section V)*.

Once we have solved the non-linear static problem in a coarse mesh, we perform an interpolation of the results in a highly refined mesh of the same domain Ω_0 , in which we will solve the linear dynamic problem. For this last step, we apply the numerical scheme presented in Merlini et al. (2025), that allows to account for the complexity of the material and overcome limitations related to the

Table 1

Material parameters for isotropic (μ) and anisotropic behavior (k_1 and k_2), mass density ρ and penalization constant for mixed formulation λ .

μ [MPa]	k_1 [MPa]	k_2	ρ [g/mm ³]	λ [MPa]
0.05	0.44	0.02	10^{-3}	50

enforcement of incompressibility. The method is fully explicit, energy-preserving and second-order accurate. The space discretization is based on high-order spectral elements, mass-lumping together with Gauss–Lobatto quadrature rule and an inf–sup stable mixed formulation. The stability of the scheme is strongly reduced by the enforcement of incompressibility. We proposed a relaxation of the CFL condition through the use of Chebyshev polynomials, thus allowing us to increase the efficiency of the computations.

We underline that the problem could have been solved with a unique solver based on total or updated Lagrangian formulation to first pressurize the cornea as a static problem, then restart the calculation as an elastodynamics problem to apply the perturbation. However, the different dynamics of the two problems coupled with the enforcement of incompressibility requires the development of two different solvers, especially for the elastic wave propagation problem. Indeed, this last solver (Merlini et al., 2025) allows obtaining accurate simulations (with more than 2 million degrees of freedom) of wave propagation in the inflated cornea comprised with the full model, in less than an hour on a normal computer. The speed and the accuracy are crucial aspects to make the model applicable in a clinical context.

The two solvers have been developed within an internal software and the resulting simulations are shown here with Paraview.

4.2. Material and geometrical parameters of the cornea model

To model the mechanical response of the cornea, we consider the material parameters given in Table 1. These parameters have been chosen based on the work of Giraudet et al. (2022), so that the apex displacement is comparable to the experimental observations of the literature.

The shape of the cornea here is derived from the analytical model (Pandolfi and Holzapfel, 2008) describing the cornea surface as an ellipsoid. For the sake of accuracy, we remark that the geometrical model is obtained as an approximation of the cornea in physiological condition, such that an iterative process should be followed to find the geometry corresponding to the actual reference configuration (Pandolfi and Holzapfel, 2008). Here, we make the simplifying assumption that the geometrical model yields the reference configuration.

The efficiency of the scheme is also conditioned by the minimum mesh element size, such that inhomogeneous mesh can strongly reduce the efficiency of the computation. Therefore, we have developed a quasi-homogeneous triangularization of a circle constructed with invertible functions, that maps the *reference element* with the element of the cornea mesh in the reference configuration $\hat{\Omega}$. This mapping is coupled with local computations, for which we do not assemble any matrix.

Once we are able to construct the circular mesh, the extension to a 3D cylindrical mesh is trivial. We derive the deformation map ϕ_c from a cylinder to the cornea first stretching the cylinder to have a closer shape from the human cornea in plane: we obtain an oval shape with the principal radius in the naso-temporal and superior–inferior directions. For the "out-of-plane" component, we consider a thickness 500 μ m (apex thickness) and we obtain the final corneal shape by applying the classical biconic function (Pandolfi and Holzapfel, 2008). All the details are presented in *Supplementary material (section VI)*.

With this final mapping ϕ_c we obtain the reference configuration $\hat{\Omega}$ of the cornea. The mapping ϕ_c is also used for the definition of the set of directions $\hat{\mathbf{r}}(\xi, \theta)$ within the anisotropic law, such that

$$\hat{\mathbf{r}}(\theta) = \nabla \phi_c \cdot \mathbf{M}(\theta) \quad \text{with} \quad \mathbf{M}(\theta) = [\cos(\theta), \sin(\theta), 0.] \quad (23)$$

We obtain that $\hat{\mathbf{r}}$ defines a parametrized set of directions lying in the plane tangent to the surface of the cornea at point ξ .

4.3. Transient and spatial distributed source

The transient volume source f used to excite the propagation of the shear wave is given by the product between a C^∞ function with a compact support for the spatial distribution of the source, and a Ricker wavelet for the excitation in time. The source term is defined as follows

$$f(\mathbf{x}, t) = \begin{cases} A_m \cdot \left[\left(\frac{t-t_0}{\sigma_\tau} \right)^2 - \frac{1}{\sigma_\tau} \right] \cdot \exp \left[\frac{1}{2\sigma_s} \left(\frac{r_0^2}{d^2 - r_0^2} + 1 \right) \right] \exp \left[-\frac{(t-t_0)^2}{\sigma_\tau} \right] & \text{if } d^2 < r_0^2 \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

where A_m is the maximum amplitude of the source, localized at point $\mathbf{x}_c$. The maximum spatial width of the source is denoted with r_0 and the distance from the center is given by $d = \sqrt{\sum_{i=1}^{dim} (x_i - x_{c,i})^2}$. The source is applied with a small offset t_0 , while σ_s and σ_τ denotes the spatial and temporal covariance. We use for the source the parameters of Table 2. In all our simulation, the source is localized at the apex of the cornea in the middle of its thickness.

Table 2

Sources parameters: A_m max amplitude, r_0 the spatial width, t_0 the offset in time, and the covariance in space σ_s and time σ_τ .

A_m [N s ² /mm ³]	r_0 [mm]	t_0 [ms]	σ_s	σ_τ [ms ²]
1.0	0.05	0.01	0.5	10 ⁻³

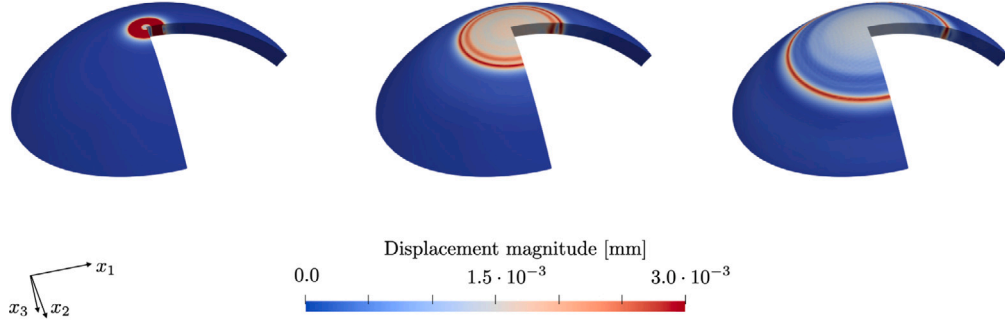


Fig. 3. 3D simulation of elastic wave propagating in cornea at rest (without intra-ocular pressure) with isotropic behavior at times [0.1, 0.3, 0.5] ms. The figure shows the magnitude $||\tilde{u}||$ of the displacement.

5. Results: geometry and material anisotropy

As a first case, we observe the propagation of the elastic wave in the solid at rest, in order to only consider the effect of the material model over the propagating wave, keeping for later the analysis of the prestress effect given by the IOP.

At rest $P_0 = 0$, we have no difference between the domains $\hat{\Omega} = \Omega_0 \approx \Omega$. Furthermore, the elastodynamic problem presented in (4) is simplified, as the geometric stiffness term cancels out ($\sigma_0 = 0$). The material behavior is only given by the linearized material stiffness term. The volumetric stiffness is unchanged; the isotropic component of the shear reduces to

$$\mathbb{C}^{iso} : \varepsilon(\tilde{u}) = 2\mu\varepsilon(\tilde{u}), \quad (25)$$

while the anisotropic component is:

$$\begin{cases} \mathbb{C}^{ani} : \varepsilon(\tilde{u}) = \frac{1}{\pi} \int_0^\pi k_1 \rho(\xi, \theta) \tilde{\mathbb{R}}(\theta) d\theta : \varepsilon(\tilde{u}) \\ \tilde{\mathbb{R}}(\theta) = \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \otimes \mathbf{r}(\theta) \end{cases} \quad (26)$$

The evaluation of the anisotropic contribution involves calculating the integral in θ for all the degrees of freedom of our finite element space. As mentioned before, we use high order spectral elements with highly refined mesh necessary for wave propagation problem, such that the total number of DoFs in elastic wave simulation for the cornea is in the order of few millions. This highlights the necessity of optimizing the calculation of the integral in θ . Given the regularity and periodicity of the function $g(\theta) = \rho(\xi, \theta) \tilde{\mathbb{R}}(\theta)$, the trapezoidal rule is the optimal choice for the approximation of the integral, with high speed of convergence (Rahman and Schmeisser, 1990). We then use $N_i = 6$ integration points equally distributed in the set $[0, \pi - \pi/N_i]$ together with the trapezoidal rule.

In the following section, we would like to compare qualitatively the resulting displacement from the *isotropic model* with $\mathbb{C} = \mathbb{C}^{iso}$ (25) and from the *anisotropic model* with the additional term of the material stiffness given in (26), such that $\mathbb{C} = \mathbb{C}^{iso} + \mathbb{C}^{ani}$. The aim is to understand what is the effect of the introduced in-plane fiber distribution over the propagating wave, which will modify the material stiffness.

5.1. Results for cornea geometry

Fig. 3 shows snapshots of the elastic wave propagation in a 3D cornea with a linear isotropic behavior (without the anisotropic components), at times $t = \{0.1, 0.3, 0.5\}$ ms. We observe a circular wave, reflecting the symmetries of the problem (in geometry, material and source).

As a first step, we observe the effects on the wave propagation due to the anisotropy and the material stiffness introduced by the fibers. We show in Figs. 4–7 the juxtaposition of results from the isotropic (left) and anisotropic (right) model. We present simulations on only one half of the domain with respect to axis x_2 , considering symmetries of geometry, source and material. From the resulting displacement field $\tilde{u}$, we extract the component $\tilde{u}_3$, shown in Fig. 4, and the $\tilde{u}_2$ component in Fig. 5. OCE techniques give usually have information only of the x_3 -component of the solution. Although we have information about the projection of $\tilde{u}$ on

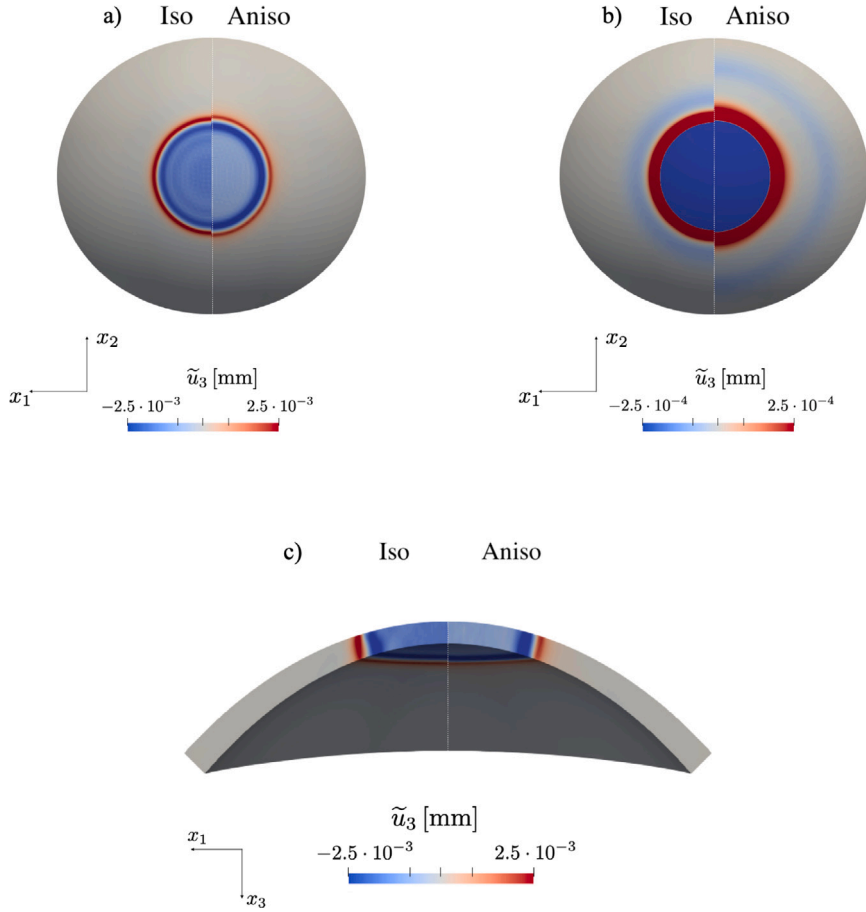


Fig. 4. 3D simulation of elastic wave propagating in cornea at rest. The figures show the $\tilde{u}_3$ component of the displacement at $t = 0.3$ ms. Left half of each figure corresponds to the simulation with isotropic model, while right half to the simulation with anisotropic model. (a) Top view of the cornea. The color bar shows a displacement in the range $[-2.5, 2.5] \mu\text{m}$; The wave is slightly faster in the anisotropic model. (b) Same solution but with a color bar with a smaller range $[-0.25, 0.25] \mu\text{m}$. One can see a second, faster, wave which is faster and slightly sensitive to the fibers in the anisotropic case. (c) Cross-section of the cornea with a color bar for displacements in range $[-2.5, 2.5] \mu\text{m}$.

the global Cartesian system $\mathbf{x}$, and not on a local coordinate system of the cornea, the curvature of the cornea is quite moderate, so that we will refer to $\tilde{u}_3$ and $\tilde{u}_2$ (or $\tilde{u}_1$) as the "out-of-plane" and the "in-plane" component, respectively.

In Fig. 4, we compare the $\tilde{u}_3$ component of the solution at time $t = 0.3$ ms, when the shear wave is still within the optical region. It shows that at first order (see Fig. 4(a)) the presence of the fibers has a weak effect with a slightly faster wave, although fiber stiffness parameter k_1 is almost an order of magnitude greater than the isotropic stiffness μ . The directional effect of the collagen distribution can be appreciated only as a second order effect: In Fig. 4(b), we observe the presence of a second wave in the anisotropic model (right half) that is faster in the principal directions of the fiber distribution, but we had to decrease the range of the colorbar by ten. We have observed that the speed of this wave increases with the parameter k_1 , especially in the fibers directions (naso-temporal and superior-inferior). However, the amplitude of this "anisotropic" wave is very low with respect to the central "isotropic" wave. This will make it hard to detect in a realistic experimental set. In Fig. 4(c), we have the $\tilde{u}_3$ component in the cross-section of both models. The difference between the two models is hardly visible, both being quite homogeneous through the thickness.

The in-plane component $\tilde{u}_2$ of the propagating wave presents similar characteristics, with a lower amplitude with respect to the out-of-plane $\tilde{u}_3$ displacement. The presence of the fibers is still slightly visible at first order in Fig. 5(a), while at second order (see Fig. 5(b)) we clearly see a faster low amplitude wave sensitive to fibers stiffness and orientation. Through the cross-section $x_1 - x_3$ (see Fig. 5(c)), we show the non-zero in-plane component $\tilde{u}_1$, less homogeneous through the thickness for the isotropic model.

We remark that the solution is given in Cartesian coordinates, such that the components of the solution we are looking at are not exactly the in-plane or out-of-plane component of the wave. In order to distinguish the effect of the material model from the curvature of the cornea, we then consider an equivalent problem with a plate geometry. The simplified geometry may help us to understand how the material properties affect the resulting response of the tissue.

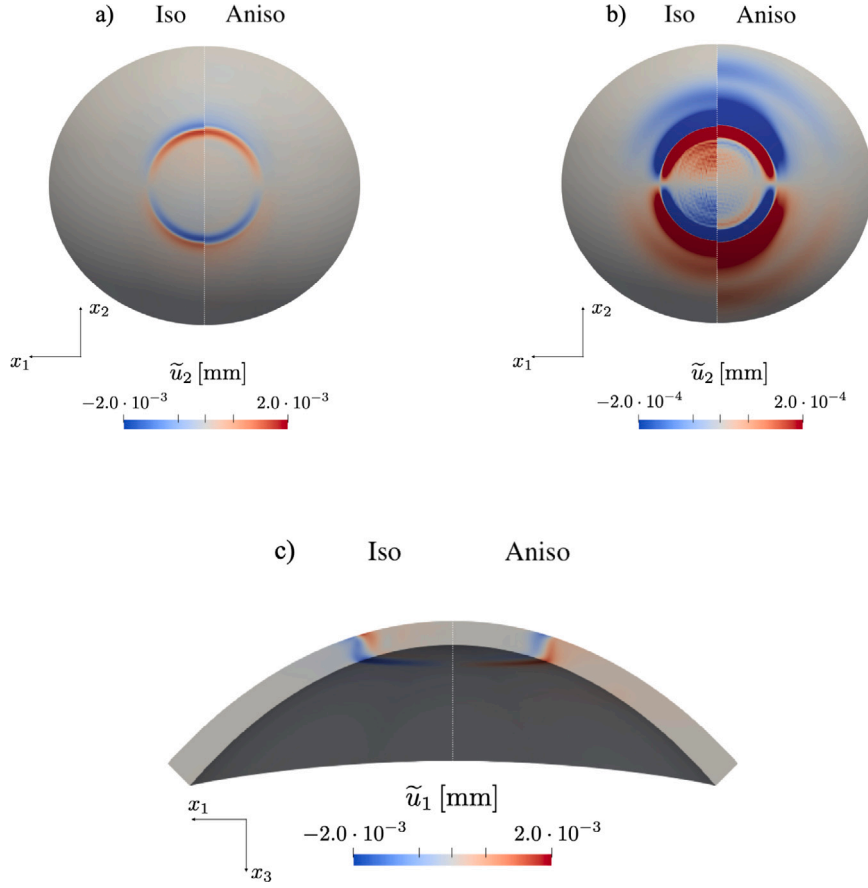


Fig. 5. 3D simulation of elastic wave propagating in cornea at rest. The figures show the in-plane components ($\tilde{u}_2$ or $\tilde{u}_1$) of the displacement at $t = 0.3$ ms. Left half of each figure corresponds to the simulation with the isotropic model, while right half to the simulation with the anisotropic model. (a) Top view of the cornea showing $\tilde{u}_2$ displacement in range $[-2.0, 2.0] \mu\text{m}$; the wave is slightly faster in the anisotropic model. (b) Same solution but with a color bar with a smaller range $[-0.2, 0.2] \mu\text{m}$. As in Fig. 4, the anisotropic model shows a faster wave of low amplitude sensitive to the presence of fibers. (c) $\tilde{u}_1$ displacement in the cross-section, for displacements in range $[-2.0, 2.0] \mu\text{m}$.

5.2. Results for a plate geometry

For the following simulations we consider a square plate of thickness 0.5 mm and edge of 10 mm, such that the dimensions are similar to those of the cornea. We localize the source at the center of the plate. All the other parameters remains the same. We compare again the results from the isotropic and anisotropic models.

Results are quite similar to what we had with the cornea without prestress. For both out-of-plane (Fig. 6) and in-plane (Fig. 7) components, the addition of the anisotropy leads to a slightly faster central shear wave with a lower amplitude with respect to the isotropic case. As for the cornea, the anisotropic model exhibits the presence of a wave with higher velocity with respect to the central one, sensitive to fibers orientation and stiffness, but with a much lower amplitude.

With these results, we conclude that our observations on the cornea geometry are not linked to its shape but only to the material parameters. The only effect of the curvature is seen on the cross-section (see Fig. 7(c)): for the plate, the in-plane component is a surface wave, with almost zero amplitude through the thickness, while in the cornea we have a non-zero component of $\tilde{u}_1$ through the thickness.

The simplified geometry allows a reduction of the problem complexity in terms of the stiffness tensor. The set of direction $\mathbf{r}$ in which we evaluate the contribution of the fibers is now given by

$$\mathbf{r} = [\cos(\theta) \quad \sin(\theta) \quad 0]^T$$

Then, the resulting fourth order tensor $\tilde{\mathbb{R}}$ has only 6 non-zero independent components. In Mandel's notation we write

$$\mathbf{r} \otimes \mathbf{r} = \begin{bmatrix} \cos(\theta)^2 & \sin(\theta)^2 & 0 & 0 & 0 & \sqrt{2} \cos(\theta) \sin(\theta) \end{bmatrix}^T,$$

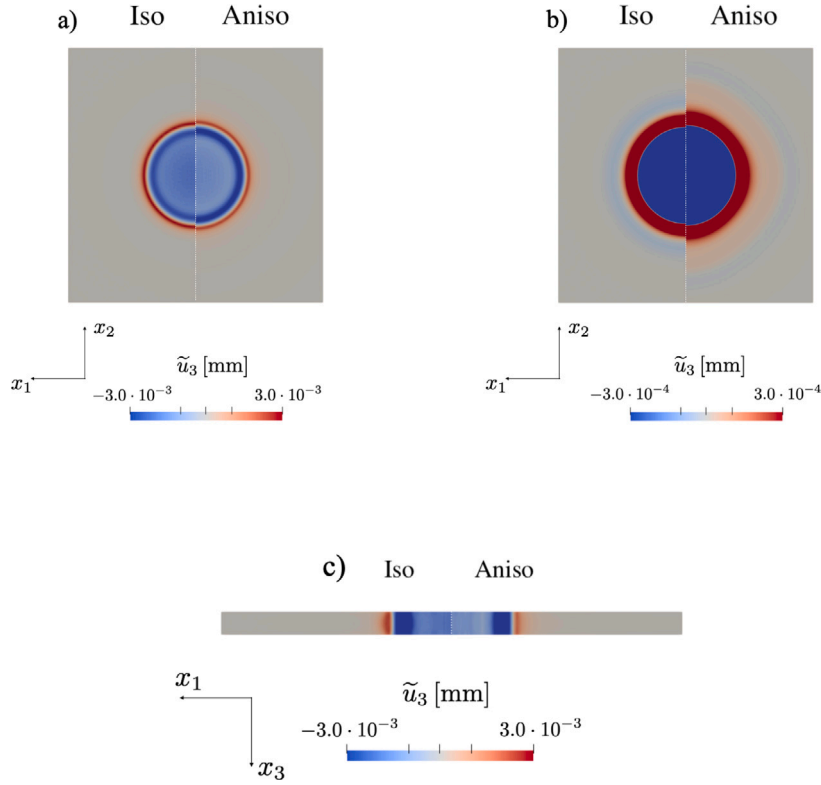


Fig. 6. 3D simulation of elastic wave propagating in plate. The figures show the $\tilde{u}_3$ component of the displacement at $t = 0.3$ ms. Left half of each figure corresponds to the simulation with the isotropic model, while right half to the simulation with anisotropic model. (a) Top view of the plate. The color bar shows a displacement in the range $[-2.5, 2.5]$ μm , with a wave in the anisotropic model slightly faster. (b) Same figure, but with a color bar with a smaller range $[-0.3, 0.3]$ μm . The anisotropic model exhibits a faster wave of low amplitude, slightly sensitive to the presence of fibers. (c) Cross-section of the cornea, with a color bar for displacements in range $[-3, 3]$ μm .

and we can easily represent the additional terms of the linearized material stiffness related to the anisotropic part. Given the symmetries of the 4-th order tensor $\mathbb{C}$, we have

$$\int_{\Omega} \tilde{\sigma} : \epsilon(\gamma) \, d\Omega = \int_{\Omega} (\mathbb{C} : \epsilon(\tilde{u})) : \epsilon(\gamma) \, d\Omega.$$

We represent the 4th-order tensor $\mathbb{C}$ using Mandel notation. We can separate the contribution of the shear and volumic terms as in Eq. (4). Focusing on the shear, the isotropic model has non-zero elastic constants only on the diagonal of $\mathbb{C}$, such that $C_{ijij} = 2\mu$ (see Fig. 8). The volumetric deformations are taken into account through the unknown pressure term $\tilde{p}$.

For the anisotropic model, we have the additional terms related to the microsphere model: $\mathbb{C} = \mathbb{C}^{iso} + \mathbb{C}^{ani}$. In particular, in the optical region we have that $\rho(\xi, \theta) = \rho_1(\theta) = \rho_1(-\theta)$ as given in relations from Eqs. (8) to (10). With this symmetry independent of the spatial position, we can further reduce the representation of the anisotropic component and derive the terms related to the anisotropic shear stiffness tensor $\mathbb{C}^{ani}$ in Mandel's notation, fully defined by

$$\begin{aligned} C_{1111}^{ani} &= \frac{1}{\pi} \int_0^\pi k_1 \rho_1(\theta) \cos^4 \theta \, d\theta \\ C_{1122}^{ani} &= C_{2211}^{ani} = \frac{1}{\pi} \int_0^\pi k_1 \rho_1(\theta) \cos^2 \theta \sin^2 \theta \, d\theta \\ C_{2222}^{ani} &= \frac{1}{\pi} \int_0^\pi k_1 \rho_1(\theta) \sin^4 \theta \, d\theta \\ C_{1212}^{ani} &= \frac{1}{\pi} \int_0^\pi 2k_1 \rho_1(\theta) \cos^2 \theta \sin^2 \theta \, d\theta \end{aligned}$$

We recall that the displacement component that is usually captured in OCE measurements corresponds to the propagating shear wave given by $\tilde{u}_3$, which is mainly driven by the components C_{2323} and C_{1313} of the elasticity tensor. In both models, as we can see in Fig. 8, these terms are given by the isotropic shear constant μ . The introduction of the fibers increases the in-plane stiffness of the material and a coupling between the in-plane component of the deformation ϵ_{11} and ϵ_{22} , but does not affect directly the

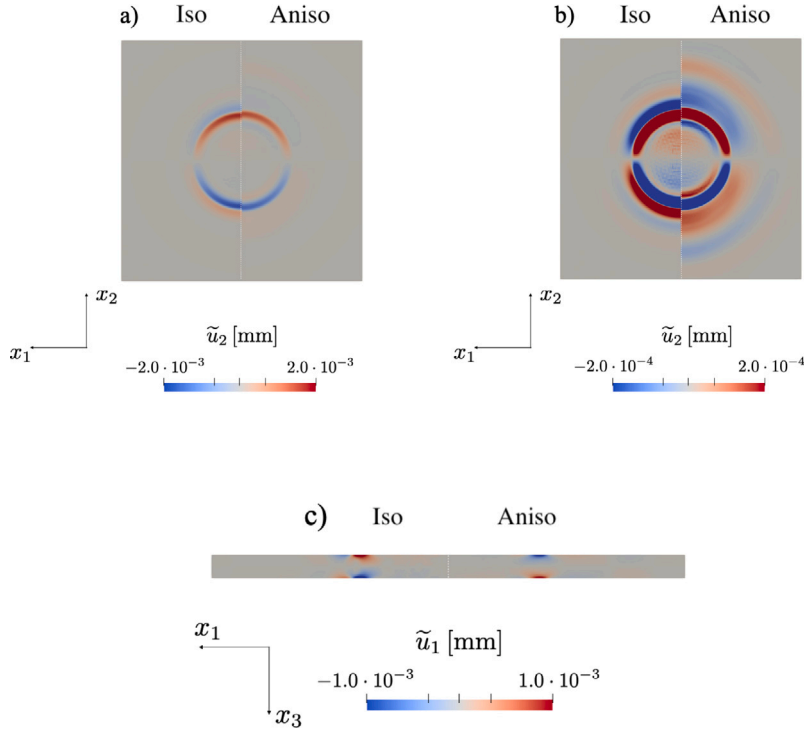


Fig. 7. 3D simulation of elastic wave propagating in plate. The figures show the in-plane components ($\tilde{u}_2$ or $\tilde{u}_1$) of the displacement at $t = 0.3$ ms. Left half of each figure corresponds to the simulation with the isotropic model, while right half to the simulation with the anisotropic model. (a) Top view of the cornea showing $\tilde{u}_2$ displacement in range $[-2.0, 2.0] \mu\text{m}$; the wave is slightly faster in the anisotropic model. (b) Same solution but with a color bar with a smaller range $[-0.2, 0.2] \mu\text{m}$. The anisotropic model shows a faster wave of low amplitude sensitive to the presence of fibers. (c) $\tilde{u}_1$ displacement in the cross-section, for displacements in range $[-1.0, 1.0] \mu\text{m}$.

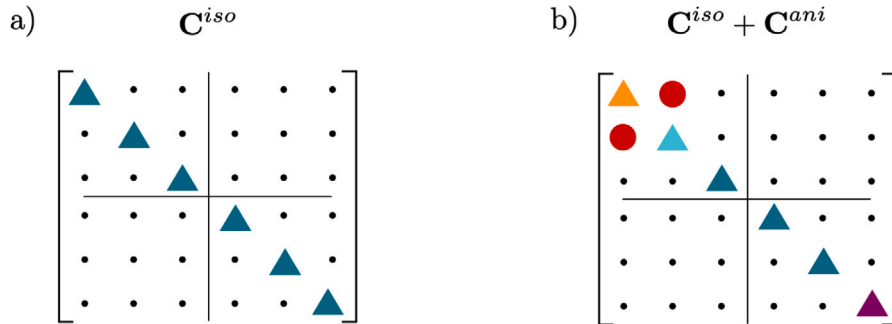


Fig. 8. Visual representation of the stiffness tensor for the isotropic and anisotropic models, for a plate. Blue triangles correspond to the same value in both tensors (2μ). Black dots denote null value.

elastic constants regarding the out-of-plane component $\tilde{u}_3$. This might explain why the introduction of the fibers (in-plane) cannot be captured at first order through the main wave.

One may notice that the analysis of waves may be associated to the fact that the cornea (and the plate) are rather thin structure (Achenbach, 2012). The solution that we obtain could be understood through a waveguide analysis — even if such analysis for a curved, nearly-incompressible, cornea is not straight forward. This may explain why we observe an out-of-plane, low amplitude, wave sensitive to the anisotropy even if the fibers are only in-plane: the thin structure creates a coupling between in-plane waves and out-of-plane shear waves.

6. Effect of prestress: a preliminary analysis on a sphere

The computation of the full problem of the cornea subjected to the intraocular pressure is not trivial to solve. Then, the main problem is to distinguish the contribution of the different elements to the dynamic response of the tissue. It is therefore interesting to

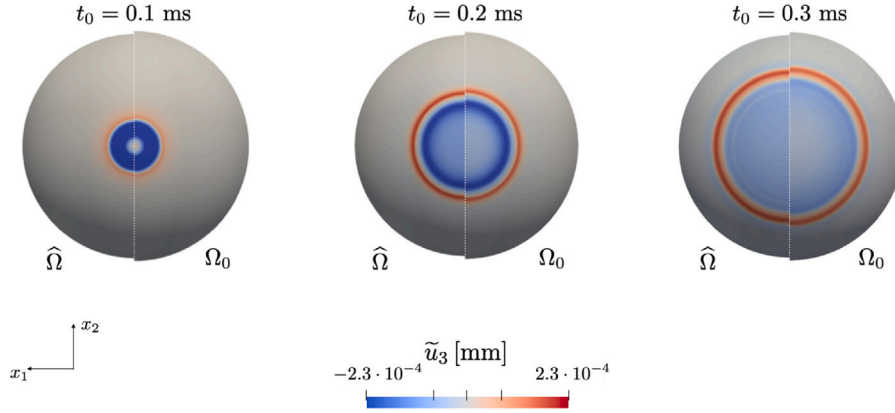


Fig. 9. 3D simulation of elastic wave propagating in an incompressible sphere. The figures show the $\tilde{u}_3$ component of the displacement at different times (From left to right: wave propagation at times [0.1, 0.2, 0.3] ms). Left half of each figure corresponds to the simulation in the configuration at rest, right half to the simulation in the configuration with an applied pressure $P_0 = 3$ kPa.

start by approximating the cornea as a portion of a thick hollow sphere inflated by an inner pressure, which allows a semi-analytical resolution of the non-linear static problem. The treatment of this static problem is classical in mechanics, and we do not present it here, but it is detailed in *Supplementary material (section VII)*.

For the wave propagation (Eq. (3)), the linearization of the shear material stiffness is obtained as for the one cornea. The only difference is that the material here is incompressible, such that the linearization is obtained with respect to the solution given by the displacement $\mathbf{u}$ and the pressure p field, for which we consider the additional perturbation

$$p = p_0 + \delta \tilde{p}.$$

The linearized material stiffness is then written as

$$\int_{\Omega_0} \boldsymbol{\epsilon}_{\mathbf{x}_0} : \mathbb{C} : \boldsymbol{\epsilon}_{\mathbf{x}_0} d\Omega_0 = \int_{\Omega_0} \zeta(\mathbf{x}_0) \boldsymbol{\epsilon}_{\mathbf{x}_0} : \boldsymbol{\epsilon}_{\mathbf{x}_0} d\Omega_0,$$

with

$$\zeta(\mathbf{x}_0) = \frac{2}{3} \mu \operatorname{tr}(\mathbf{C}_0) + 2p_0.$$

The geometric stiffness is given by the stress tensor $\boldsymbol{\sigma}_0$ rewritten in Cartesian coordinates $\mathbf{x}_0$.

In the following simulations, we consider only the isotropic material model with the parameters of the cornea (see Table 1). We simulate only half a sphere for simplicity. The inner pressure is $P_0 = 3$ kPa (equivalent to 22.5mmHg); the excitation source is the same as the one of the cornea, placed at the apex. The hexahedral mesh is discretized with 2.5M DoFs, spectral elements of fourth order for the displacement field and discontinuous second-order elements for the pressure contribution.

We compare the propagation of the wave for the sphere with and without inner pressure. The Fig. 9 shows the $\tilde{u}_3$ -component in the spherical approximation Ω_0 of pressurized sphere (right half) with the shear wave propagating in the sphere “at rest” $\hat{\Omega}$ (left half). Note that the sphere under pressure is slightly bigger than the one without pressure. The wave in the preloaded configuration Ω_0 propagates faster, even if the sphere is larger. These results underline the importance of accounting for the non-linear static problem in order to have a correct reproduction of shear wave elastographic measurements.

Furthermore, we would like to distinguish the contributions of the material and geometric stiffnesses on the wave propagation. We then compare the solution of the full linear problem with two modified linear problems: a first one without the prestress $\boldsymbol{\sigma}_0$, in order to account only for the material stiffness, and a second with a material stiffness linearized around rest condition, to account only for the geometric stiffness. Keep in mind that these two modified problems are simulated around the apparent configuration Ω_0 . Results are presented in Fig. 10. We observe that the linearized material behavior generates a propagating shear wave that is slower and with larger amplitude than the actual $\tilde{\mathbf{u}}$. This result is coherent with the present choice of the material law, as the Neo-hookean material exhibits a decreasing stiffness under increasing load. On the contrary, the simulation based only on the geometric stiffness with a prestress $\boldsymbol{\sigma}_0$ resulting from the non-linear static problem is homogeneously faster than the actual solution of the full elastodynamic problem. This result can be attributed to the circumferential component of the prestress $\sigma_{0,\theta\theta}$ that is positive through the thickness and $\sigma_{0,\theta\theta} > |\sigma_{0,rr}|$ (with $\sigma_{0,rr}$ the negative radial component of the prestress), which makes the sphere geometrically stiffer through the thickness, with an isotropic increase in the wave propagation velocity.

Notice that in a complete model of the wave propagation problem, we should account for the exponential law of collagen fibers, such that both material and geometric stiffnesses increase with the preload. With these results, we conclude that an accurate evaluation of the preload and a proper description of the material behavior have non-negligible impact on the reproduction of wave propagation measurements for the cornea.

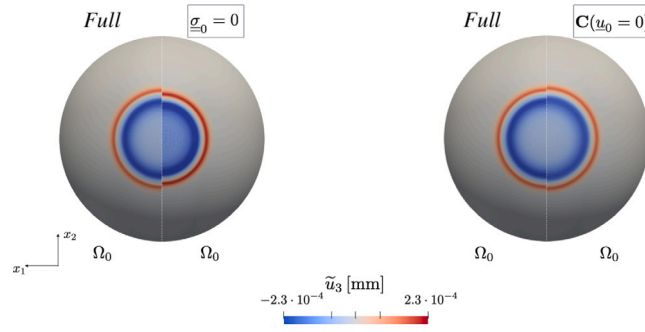


Fig. 10. 3D simulation of elastic wave propagating in an inflated incompressible sphere, with $P_0 = 3$ kPa. The figures show the $\tilde{u}_3$ component of the displacement at $t = 0.2$ ms in the apparent configuration Ω_0 . (Left) Effect of the geometrical stiffness; Left half of the figure shows the full simulation, while right half shows solution of a modified linear wave problem without prestress (i.e. $\sigma_0 = 0$) - it includes only the contribution of the linearized material stiffness. (Right) Effect of the material stiffness; Left half of the figure shows the full simulation, while right half shows solution of a modified linear wave problem without modification of the linearized material stiffness — it includes only the effect of the prestress on the geometrical stiffness.

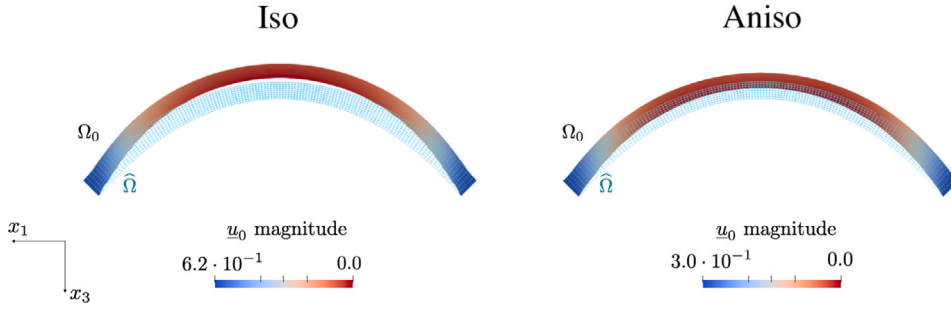


Fig. 11. Numerical results for static inflation of cornea model. Both images represent a cross-section of the inflated geometry (Ω_0) with respect to the reference domain $\hat{\Omega}$, represented here with the mesh. The color shows the x_3 -component of the static solution u_0 . (Left) Solution of the isotropic model; (Right) Solution of the anisotropic model.

7. Wave propagation with the full cornea model

We now simulate the full problem of the cornea. The first step is to compute the non-linear static problem: we apply an intraocular pressure of $P_0 = 3$ kPa. As expected, the anisotropic material exhibits a stiffer response, with a maximum displacement that is almost the half with respect to the isotropic model (see Fig. 11). The simulations were obtained with Q_2 elements and 335,259 degrees of freedom, with the same parameters as in previous simulations.

Once we have computed solution u_0 and the associated deformation gradient F_0 , we can derive the full material behavior given by the linearized material stiffness (Eq. (13)) and the geometric stiffness (Eq. (18)) for the isotropic, volumetric and anisotropic components in the apparent configuration Ω_0 of the cornea.

In the following results we compare different simulations, to put in evidence different aspects of the isotropic and anisotropic models over the propagating wave. We proceed as before to compare the models, given the symmetry of the source, geometry and material: the representation of the cornea is split in two, where each half representing a different situation. The configurations now are not the same for the isotropic and anisotropic models, since the computed domain from the non-linear static problem Ω_0 is not the same for a different material behavior, so that we will distinguish them with notations Ω^{iso} and Ω^{ani} . We need to clarify that the Dirichlet boundary condition prevents a change in radius of the cornea, and thus the two configurations are not distinguishable from the top view.

In Fig. 12, we compare the elastic wave propagating in the simplest model given by the cornea with no IOP applied ($\Omega_0^{iso} = \hat{\Omega}$) and isotropic behavior (left half of each cornea in Fig. 12) with the complete model, characterized by the Ω_0 obtained from the resolution of the anisotropic non-linear static problem with $P_0 = 3$ kPa (Ω_0^{ani}) on the posterior surface $\hat{\Gamma}_i$ and the linearized material behavior including material (Eqs. (14), (15) and (16)) and geometric (Eqs. (19), (20) and (21)) stiffness. On the left we present the $\tilde{u}_3$ component, and on the right $\tilde{u}_2$ component of elastic wave propagating in the top surface the cornea. The wave propagating in the prestressed anisotropic configuration clearly travels faster with respect to isotropic wave without the intraocular pressure, as we can see for both components. In addition, the directional effect of the fibers still remains a second order effect, although their contribution should be more relevant considering the associated exponential law given in Eq. (7). We recall the results shown in

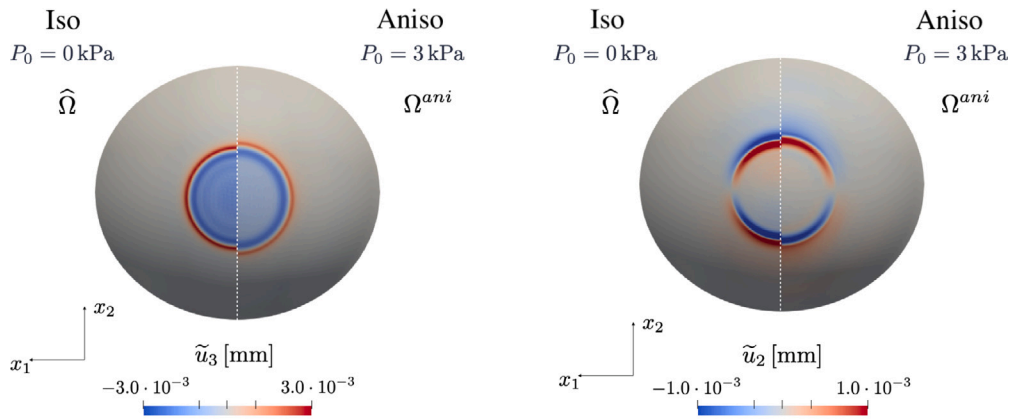


Fig. 12. 3D simulation of elastic wave propagating in cornea at $t = 0.3$ ms. Left half of each figure corresponds to the unloaded isotropic model (so to the reference configuration $\hat{\Omega}$), while right half corresponds to loaded anisotropic model. (Left) out-of-plane component of the solution ($\tilde{u}_3$). (Right) in-plane component of the solution ($\tilde{u}_2$). In both figures, the wave travels faster in the loaded anisotropic model, while no directional effect related to fiber distribution is observed.

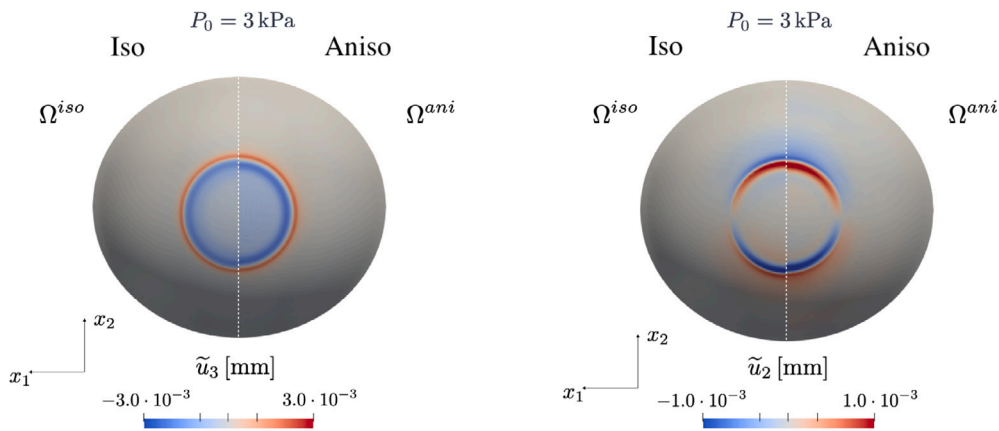


Fig. 13. 3D simulation of elastic wave propagating in cornea under pressure $P_0 = 3$ kPa, at $t = 0.3$ ms. Left half of each figure corresponds to the loaded isotropic model, while right half corresponds to loaded anisotropic model (so, the same as in Fig. 12). (Left) out-of-plane component of the solution ($\tilde{u}_3$). (Right) in-plane component of the solution ($\tilde{u}_2$). In both figures, the wave travels almost at the same speed, with the isotropic wave slightly faster.

Figs. 4 and 5, in which without the prestress the effect of the fibers was slightly visible at a lower scale and mainly with the in-plane component $\tilde{u}_2$. Present results show the relevance of the prestress in order to properly simulate dynamic OCE measurements.

As a further step, we investigate the elastodynamic response of both isotropic and anisotropic model linearized around $\mathbf{u}_0$, given by the applied intraocular pressure $P_0 = 3$ kPa. The domains in which the elastic wave propagates correspond to those obtained from the non-linear static problem (Fig. 11), with the respective material model. A comparison of the resulting displacements of the two models is given in Fig. 13. We show on the left the $\tilde{u}_3$ component of the solution and on the right the $\tilde{u}_2$ component. The elastic wave in the preloaded isotropic solid (left half of each cornea) propagates with almost same speed and wavefront of the anisotropic model (right half of each cornea — same results shown in Fig. 12). As we have shown for the unloaded simulations of the cornea (see Section 5.1), the principal shear wave is not affected by the directional stiffness introduced with the microsphere model. We again observe an additional wave that is faster in the fiber direction (symmetric results are obtained for the x-component), revealing the anisotropic effect of the collagen fibers, but with a much lower amplitude.

To summarize, we do not observe a significant difference on the propagating shear wave between the isotropic and anisotropic shear model at first order. The prestress has a relevant impact on the speed of the propagating wave, which is instead much less sensitive to the in-plane distribution of the collagen fiber. These results are coherent with what we found from the study of the cornea in the unloaded configuration (with no intraocular pressure) and the study conducted over the sphere. The inclusion of fibers contribution (in-plane) within the model is relevant only for the inflation problem, as the material exhibits a stiffer behavior and the maximum displacement is almost the half of the one obtained with the isotropic model. The prestress related to the collagen contribution might still have an effect in-plane, as observed for the plate, which is not affecting the propagation of the shear wave.

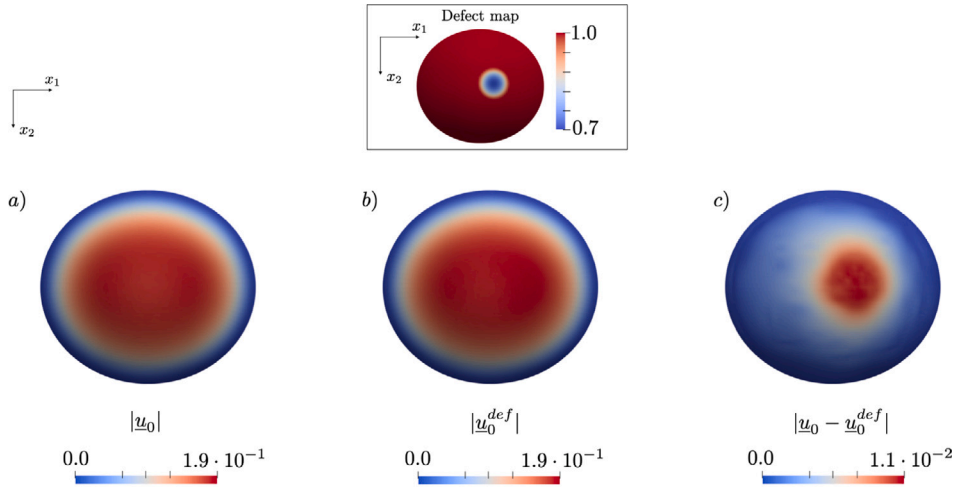


Fig. 14. Numerical results for static inflation of cornea model with $P_0 = 2$ kPa, with a defect. The color shows the magnitude of the displacement ($|u_0|$), resulting from non-linear static inflation of the cornea. The defect amplitude is given in the top box. (a) Solution for the anisotropic model without defect (same solution is in Fig. 11). (b) Solution for the anisotropic model with defect on the mechanical parameters k_1 (anisotropic) and μ (isotropic) - the defect map shows the multiplicative factor applied locally to these parameters. (c) Difference (absolute) between the displacement in the solution without (a) and with defect (b).

We conclude then that the anisotropic model could be well approximated by an isotropic behavior with a more complex law than the Neo-Hookean potential Eq. (5), like a stiffening law with the increasing stretch of the solid. A further consideration concerns the possible introduction of fibers distribution also in the out-of-plane direction, in order to account for the intertwined lamellae in the anterior part of the cornea. This could generate a coupling effect between the depth-dependent and the in-plane distribution, making the wave sensitive to the presence of fibers.

8. Wave propagation in the cornea with a defect

The variation of mechanical properties of the cornea seems to be a relevant indicator for the detection of pathologies at early stage. The principal example for the cornea is the keratoconus that consists in a progressive thinning and deformation of the cornea into a conical shape that compromises vision (Meek and Boote, 2009). The causes of this condition are still unknown. However, given the structure and the physiology of the cornea, variations of the shape are believed to be induced by a weakening of the mechanical properties of the tissue (Giraudet et al., 2022).

Here we would like to test the sensitivity of the model to a local variation of mechanical properties, and to investigate which properties of the model are the most relevant for the dynamic response of the tissue through a qualitative mechanical contrast.

The keratoconus is usually associated to a weakening of the collagen network, such that a variation of parameters stiffness for both isotropic and anisotropic components of the model are considered. We recall, in fact, that the contribution of collagen fibers is also considered within the isotropic part of the material law, to account for randomly distributed fibers. Let us then consider a reduction of mechanical stiffness according to the following relation

$$k_{defect} = k \cdot \begin{cases} 1 - 0.3 \cdot \exp \left[\frac{1}{2\sigma_s} \left(\frac{r_0^2}{d^2 - r_0^2} + 1 \right) \right] & \text{if } d^2 < r_0^2 \\ 1 & \text{otherwise} \end{cases} \quad (27)$$

with $k = \{\mu, k_1\}$ for respective isotropic and anisotropic components. As for the source Eq. (24), the term r_0 denotes the spatial width of the distribution with a covariance σ_s , while d represents the distance from the center of the mechanical weakness. In the following results, we use a distribution homogeneous through the thickness, with the defect localized in the arbitrary point $\xi_c = (1.14, -0.23, 0.43)$ mm within the optical region, with covariance $\sigma_s = 0.5$ and the spatial width $r_0 = 1.5$ mm (see Fig. 14).

We then solve the full problem starting as before from the non-linear static problem, in which we consider the reduced material parameters and an intraocular pressure applied of 2 kPa (15 mmHg) in the physiological range. We compare the displacement obtained for a 'healthy' cornea with no defect, as shown in Fig. 14(a) to the displacement of the cornea with the defect (Fig. 14(b)). The effect of the mechanical weakness on the resulting non-linear static problem is not visible on the anterior surface, since the difference between the two results (Fig. 14(c)) is one order of magnitude lower than the static solution. This could be for example a keratoconus at an early stage.

On the contrary, the propagating wave is strongly affected by the presence of the weakness. In the following results we consider the elastic wave propagation problem in the weakened cornea Ω_0 subjected to the intraocular pressure $P_0 = 2$ kPa (given in Fig. 14b).

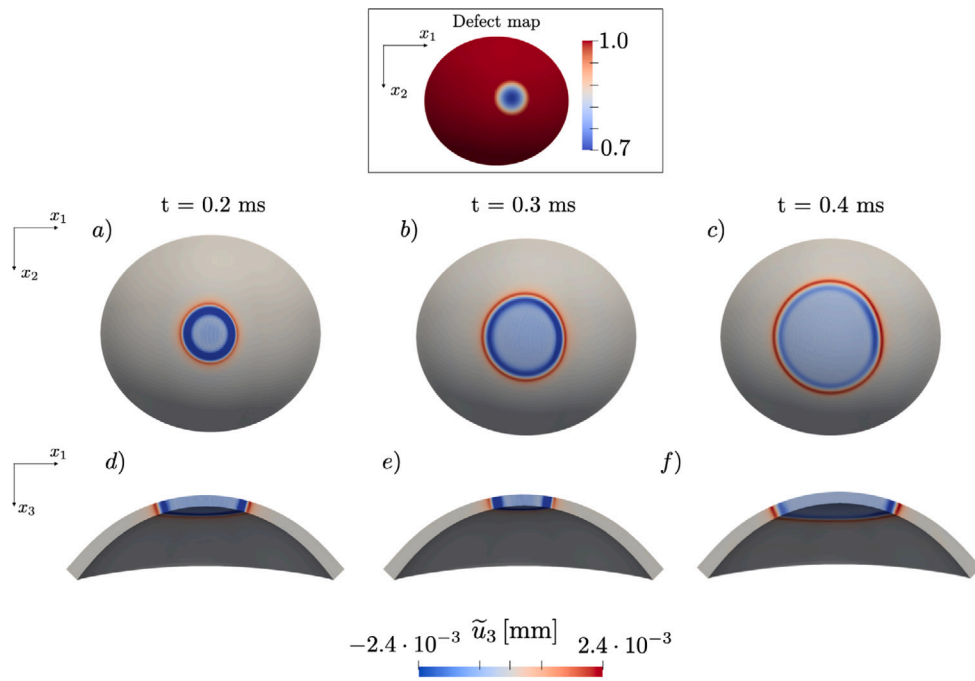


Fig. 15. 3D simulation of elastic wave propagating in cornea under pressure ($P_0 = 2$ kPa), with a defect. The color shows the out-of-plane component of the solution ($\tilde{u}_3$). The defect amplitude is given in the top box. The defect is applied on the mechanical parameters k_1 (anisotropic) and μ (isotropic); the defect map shows the multiplicative factor applied locally to these parameters. (Top row - a, b, c) Top view of the solution at different times ($t = 0.2$ ms, $t = 0.3$ ms, and $t = 0.4$ ms). (Bottom row - d, e, f) Cross-section of the solution at different times ($t = 0.2$ ms, $t = 0.3$ ms, and $t = 0.4$ ms).

In Fig. 15, we show the wave ($\tilde{u}_3$ component) propagating on the top surface of the cornea (first row) and through the thickness (second row). The front wave is strongly deformed now, due to a speed reduction of the wave at the weakened area, with a larger amplitude, clearly denoting the reduction in stiffness of the material.

Our last analysis accounts for the comparison between a mechanical weakening of the isotropic stiffness μ and the anisotropic parameter k_1 . In Fig. 16, we show the $\tilde{u}_3$ component on the top surface of the cornea considering on the first row a reduction in the stiffness of the fibers through the parameter k_1 , and on the second row a reduction of the isotropic stiffness μ . The shear wave is weakly affected by the variation of the fibers, while the impact of the isotropic term is evident.

These results suggest that if the collagen network presents only an in-plane distribution, as modeled in this work, a weakening of fibers cannot be detected with shear waves. These conclusions are overall coherent with what we have found on the previous results related to the comparison between the isotropic and anisotropic model, with or without the effect of the intraocular pressure. We recall in particular the elasticity tensor of the plate (see Fig. 8), where the terms related to the shear out-of-plane stiffness (C_{2323} and C_{1313}) depend only on the mechanical parameter μ related to the isotropic component μ of the material law.

A recent work from (Pandolfi et al., 2019) proposes a simplified truss-work model to study the interaction between different components of the fibrous reinforcement within the cornea. They consider the collagen network formed by two sets of in-plane fibers, connected within each set by *inter-cross-links* (in-plane), and across the sets with *intra-cross-links* (both in-plane and out-of-plane). The mechanical response of the model to intraocular pressure show a negligible effect when locally reducing the stiffness of in-plane fibrils or inter-cross-links. On the contrary, a weakening of the transversal bonds results in a strong deformation of the modeled cornea, which develops a shape comparable to that observed experimentally in the case of keratoconus. The intra-cross-links are then responsible for the stability of element structure. In a continuum model, as ours, the stabilizing function of intra-cross-links is accomplished by the matrix, modeled with the isotropic component. We see then, that the sensibility of our wave propagation model to a deterioration of the mechanical parameters is similar to observations obtained with the truss-work model of Pandolfi et al. (2019). We can conclude that shear wave elastography is a fundamental technique for the in-vivo study of the mechanical response of the cornea. Reproducing data with our model would enable us to reconstruct the mechanical properties of the tissue and early detect a weakening of the structure, which would lead to cornea deformation and vision deterioration.

9. Conclusions and perspectives

In this article, we propose a novel model for the reproduction of transient shear wave elastographic measurements of the cornea, based on the finite element approach. The followed approach starts with the definition of a biomechanical model of the cornea,

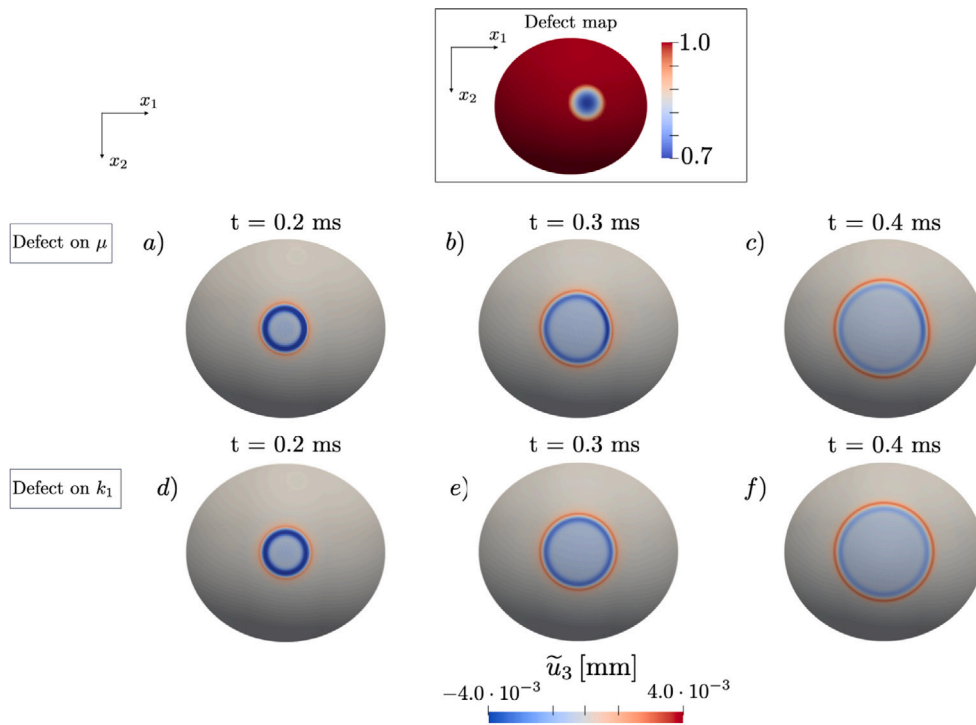


Fig. 16. 3D simulation of elastic wave propagating in cornea under pressure ($P_0 = 2$ kPa), with a defect. The color shows the out-of-plane component of the solution ($\tilde{u}_3$). The defect amplitude is given in the top box — the defect map shows the multiplicative factor applied locally to the mechanical parameters. (Top row - a, b, c) Top view of the solution for a defect on the mechanical parameters μ (isotropic) at different times ($t = 0.2$ ms, $t = 0.3$ ms, and $t = 0.4$ ms). (Bottom row - d, e, f) Top view of the solution for a defect on the mechanical parameters k_1 (anisotropic, collagen fibers) at different times ($t = 0.2$ ms, $t = 0.3$ ms, and $t = 0.4$ ms).

that encompasses nearly-incompressibility, non-linearity, anisotropy, geometry and loading conditions. Considering the elastic wave propagation as a small perturbation of the non-linear mechanics, we show how the problem is governed by a non-linear static problem that yields the configuration of the cornea in physiological conditions and a linear wave propagation problem written in the preloaded configuration. This latter formulation allows the isolation of the volumetric constraint and the application of a dedicated numerical method previously developed for elastodynamic nearly-incompressible problems.

We then consider two different numerical approaches, given the different nature of the two problems. The non-linear static formulation is solved with a classical minimization algorithm based on the non-linear conjugate gradient method, which is acceptable for moderate deformations as those related to the cornea. For the approximate wave propagation problem, the space discretization relies on the use of high-order spectral elements coupled with a highly refined mesh of the cornea geometry. The presented simulations were performed with more than 2 millions of degrees of freedom, in order to have accurate results. At the same time, the applicability of the model in a realistic scenario requires results to be obtained with an acceptable runtime. This is to emphasize how the need for accuracy clashes with the need for efficiency, making it essential to develop an efficient numerical strategy able to cope with the large size of the problem and numerical limitations associated to incompressibility. With the solver presented in [Merlini et al. \(2025\)](#), elastodynamic simulations were obtained in less than an hour on a normal computer.

The principal elements of the model are progressively analyzed through numerical experiments, considering the effect of the microsphere model, the curvature of the geometry and the prestress induced by intraocular pressure. In both at rest and preloaded conditions, simulations show that the fibers have a limited effect on the shear wave propagation, despite their high stiffness. This may be due to the present choice of in-plane fiber distribution. To be complete, we actually observe that the fibers have a strong effect on a faster wave but with a magnitude an order of magnitude lower. Thus, such wave is likely to be difficult to detect in a realistic experimental scenario. We show instead that the shear wave is highly affected by the prestress given by the intraocular pressure, first considering a simplified analytical model of an inflated thick hollow sphere and then accounting for the original problem with the full biomechanical model of the cornea.

The flexibility of the numerical approach allows to introduce a local mechanical weakening of the material in order to simulate a keratoconus at early stage. Numerical results show the sensibility of the shear wave to the presence of the material defect, proving the capability of shear wave elastography for the early detection of cornea pathologies, related to changes in the microstructure and associated weakening of mechanical properties. By comparing the effect of a defect on the isotropic and anisotropic component, results are coherent with previous observations: a defect only on the mechanical property related to the isotropic behavior clearly

deforms the front wave, while a local reduction only of fibers stiffness is not detected. From these results, we can derive different conclusions: either the in-plane anisotropy can be disregarded and consider only an isotropic potential with a stiffening behavior with stretch, either we add to the model a depth-dependent fiber distribution that couples the in-plane and out-plane stiffness of the material.

To conclude, the presented model aims to reproduce the dynamic response of a biological tissue, tackling the principal mechanical and numerical challenges of the cornea and experimental conditions. Notice that the outlined approach can be generalized for different applications that present similar mechanical conditions. Considering the specific case of the cornea, once the numerical aspects are treated and other anisotropic models are investigated, it would be possible to couple the model with d-OCE measurements, in order to obtain in-vivo characterization of the cornea and the assessment of corneal health.

CRediT authorship contribution statement

Giulia Merlini: Writing – original draft, Software, Investigation, Formal analysis. **Sebastien Imperiale:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis. **Jean-Marc Allain:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jmps.2025.106363>.

Data availability

No data was used for the research described in the article.

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